EN.553.730 Statistical Theory

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Part I

Data Reduction

0.1 Introduction

Statistical theory begins with an experiment whose outcome is random but whose distribution is only partially known. Let $(\mathcal{X}, \mathcal{A})$ be the sample space and let

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

be a family of probability measures on it. The pair $(\mathcal{X}, \mathcal{P})$, with the underlying σ-field understood, is called a *statistical experiment*. We observe $X \sim P_\theta$ and wish to learn something about the unknown parameter θ.

The raw observation often contains more detail than is relevant to this goal. A *statistic* is a measurable map

$$T : \mathcal{X} \longrightarrow \mathcal{T}.$$

Replacing X by $T(X)$ is a data reduction. Such a reduction is attractive when T is simpler than X, but simplicity alone is not enough: we would like to discard no information about θ.

Example 0.1.1 (Repeated coin flips). Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$, where $p \in [0, 1]$. The full data are the binary string $X = (X_1, \dots, X_n) \in \{0, 1\}^n$. If the order of the flips is irrelevant, a natural reduction is

$$S = \sum_{i=1}^{n} X_i,$$

the total number of heads. The sample space has been reduced from 2^n possible strings to $n + 1$ possible counts. We will see that this reduction loses no information about p.

There are two complementary ways to make "no loss of information" precise. The decision-theoretic formulation asks whether every procedure based on X can be reproduced, with the same performance, using only $T(X)$. The probabilistic formulation asks whether, after $T(X)$ is known, the remaining randomness in X has a distribution that is independent of θ. The notion of sufficiency connects these two points of view.

0.2 The Decision-Theoretic Framework

A statistical problem requires more than a model. Let $\mathcal{D}$ be an action space, and let

$$L : \Theta \times \mathcal{D} \longrightarrow [0, \infty]$$

be a loss function. The value $L(\theta, a)$ is the cost of taking action a when the true parameter is θ. A nonrandomized *decision rule* is a measurable function $\delta : \mathcal{X} \to \mathcal{D}$. Its frequentist risk is

$$R(\theta, \delta) = \mathbb{E}_\theta\Big[L\Big(\theta, \delta(X)\Big)\Big] . \tag{0.2.1}$$

Thus a rule is evaluated not by a single number, but by the function $\theta \mapsto R(\theta, \delta)$.

Common examples include the following.

- For point estimation of $g(\theta) \in \mathbb{R}$, squared-error loss is $L(\theta, a) = (a - g(\theta))^2$.

- Absolute-error loss is $L(\theta, a) = |a - g(\theta)|$.
- For testing $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$, the action space is $\{0, 1\}$, where action 1 means rejection. Zero-one loss assigns a cost to an incorrect decision; asymmetric loss may assign different costs to the two kinds of error.

Randomization is occasionally useful. Formally, a randomized rule is a Markov kernel $Q(\,\mathrm{d}a \mid x)$ from $\mathcal{X}$ to $\mathcal{D}$: after observing x, the action is drawn from $Q(\cdot \mid x)$. A deterministic rule is the special case $Q(\,\mathrm{d}a \mid x) = \delta_{\delta(x)}(\,\mathrm{d}a)$.

Suppose only $T = T(X)$ is retained. Then a rule based on the reduced data has the form $d(T)$, or more generally is a kernel from $\mathcal{T}$ to $\mathcal{D}$. The central question is whether this restriction makes any decision problem harder. If the conditional distribution of X given T does not depend on θ, a rule based on X can be simulated from T: given $T = t$, generate X^* from that conditional distribution and apply the original rule to X^*. Under every P_θ, X^* has the same marginal law as X, so the simulated rule has exactly the same risk function.

This observation motivates sufficiency. It also makes clear why randomized rules are natural in the most general decision-theoretic statement: the reconstruction of X from T may itself require randomization. For convex losses and point estimation, conditional expectation will often remove that randomization and improve risk; this is the content of the Rao–Blackwell theorem.

Example 0.2.1 (Estimating a coin bias)**.** For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ and squared-error loss, consider the rule $\delta(X) = X_1$. Conditional on $S = \sum_i X_i = s$, symmetry gives

$$\mathbb{P}_p(X_1 = 1 \mid S = s) = \frac{s}{n},$$

which does not depend on p. Hence X_1 can be simulated from S by a Bernoulli(s/n) draw. More importantly, replacing X_1 by its conditional mean S/n reduces squared-error risk. The reduction S therefore supports a strictly better nonrandomized estimator than the original rule.

0.3 Sufficiency

Definition 0.3.1 (Sufficient statistic)**.** A statistic $T = T(X)$ is *sufficient* for θ (or for the family $\mathcal{P}$) if there is a version of the conditional distribution of X given T that is the same for every $\theta \in \Theta$. Equivalently, there is a Markov kernel $K(\,\mathrm{d}x \mid t)$, not depending on θ, such that $K(\cdot \mid T)$ is a version of $P_\theta(X \in \cdot \mid T)$ for every θ.

For a discrete model this says that, whenever $P_\theta(T = t) > 0$,

$$P_\theta(X = x \mid T = t) \tag{0.3.1}$$

does not depend on θ. The general definition is phrased in terms of conditional distributions because conditioning on $T = t$ may mean conditioning on an event of probability zero.

Example 0.3.1 (The number of heads is sufficient)**.** For $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ and $S = \sum_i X_i$, every binary string x with s heads has probability $p^s(1-p)^{n-s}$. Consequently,

$$P_p(X = x \mid S = s) = \frac{p^s(1-p)^{n-s}}{\binom{n}{s} p^s(1-p)^{n-s}} = \binom{n}{s}^{-1}$$

for $\sum_i x_i = s$ (with the boundary cases interpreted on their supports). Conditional on $S = s$, the locations of the heads are uniform over the $\binom{n}{s}$ possibilities and contain no further information about p.

The conditional definition is intuitive, but direct calculation of a conditional law can be cumbersome. In dominated models the following criterion is usually easier.

Theorem 0.3.1 (Fisher–Neyman factorization theorem)**.** *Suppose $\{P_\theta : \theta \in \Theta\}$ is dominated by a σ-finite measure μ, and write $f_\theta = \mathrm{d}P_\theta / \mathrm{d}\mu$. Under the usual regularity conditions on the sample spaces, T is sufficient if and only if there are nonnegative measurable functions g_θ and h such that*

$$f_\theta(x) = g_\theta(T(x))h(x) \quad \text{for } \mu\text{-almost every } x, \tag{0.3.2}$$

where h does not depend on θ.

Proof sketch. If (0.3.2) holds, then within a fiber $\{x : T(x) = t\}$ the relative weights of possible observations are determined by $h(x)$, independently of θ. Normalizing these weights gives a parameter-free conditional law of X given $T = t$. Conversely, if this conditional law is parameter-free, disintegrate P_θ into the marginal law of T and the common conditional law of X given T; their densities provide g_θ and h. A fully general proof uses regular conditional probabilities and the Radon–Nikodym theorem. □

Example 0.3.2 (Bernoulli sample)**.** For $x \in \{0,1\}^n$,

$$f_p(x) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} = p^{\sum_i x_i}(1-p)^{n-\sum_i x_i}.$$

This factors through $S = \sum_i X_i$, with $h(x) = 1$, so S is sufficient for p. Any one-to-one transformation of S is also sufficient; for example, the sample mean $\bar{X} = S/n$ is sufficient.

Example 0.3.3 (Normal location and scale)**.** If $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, then

$$f_{\mu,\sigma^2}(x) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \left(\sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2 \right) \right\}.$$

The factorization theorem shows that $T(X) = (\sum_i X_i, \sum_i X_i^2)$ is sufficient for (μ, σ^2). When $n \geq 2$, this is equivalent to retaining $(\bar{X}, \sum_i (X_i - \bar{X})^2)$.

Sufficiency concerns the family of distributions, not the way in which the parameter is named. If T is sufficient for θ, then it is sufficient for any function $g(\theta)$. The converse need not hold if the model is changed so that only $g(\theta)$ is regarded as unknown.

0.3.1 Exponential Families

A dominated family is a k-parameter *exponential family* if its density can be written

$$f_\eta(x) = h(x) \exp\Big\{\eta^\top T(x) - A(\eta)\Big\}, \qquad \eta \in H \subseteq \mathbb{R}^k. \tag{0.3.3}$$

Here η is the natural parameter, $T(x) \in \mathbb{R}^k$ is the natural statistic, and

$$A(\eta) = \log \int h(x) e^{\eta^\top T(x)}\, \mu(\mathrm{d}x)$$

is the log-partition function. The natural parameter space is

$$H_0 = \left\{\eta \in \mathbb{R}^k : \int h(x) e^{\eta^\top T(x)}\, \mu(\mathrm{d}x) < \infty\right\}.$$

A family is called *full* when its parameter space is H_0, and *regular* when H is open. Some authors use slightly different terminology, so these conditions should always be stated explicitly.

For an iid sample with one-observation density

$$f_\eta(x) = h(x) \exp\{\eta^\top t(x) - A(\eta)\},$$

the joint density is

$$\left\{\prod_{i=1}^n h(x_i)\right\} \exp\left\{\eta^\top \sum_{i=1}^n t(x_i) - nA(\eta)\right\}.$$

Thus $\sum_i t(X_i)$ is sufficient, and its dimension does not grow with the sample size.

Example 0.3.4 (Bernoulli family in natural form)**.** For $0 < p < 1$, set $\eta = \log\{p/(1-p)\}$. Then

$$p^x(1-p)^{1-x} = \exp\{x\eta - \log(1+e^\eta)\}, \qquad x \in \{0,1\}.$$

Hence $A(\eta) = \log(1+e^\eta)$ and the natural statistic for an iid sample is $S = \sum_i X_i$. The parameter space is all of $\mathbb{R}$, so this is a full, regular, one-parameter exponential family.

Example 0.3.5 (A curved exponential family)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \mu^2)$ with $\mu > 0$. The joint density factors through $(\sum_i X_i, \sum_i X_i^2)$, but its two natural parameters are constrained by a one-dimensional relation. It is therefore a curved subfamily of a two-parameter exponential family. Curved families need not inherit all completeness properties of the containing full family.

The log-partition function encodes moments of the natural statistic. If differentiation under the integral is justified and η is in the interior of the natural parameter space, then

$$\nabla A(\eta) = \mathbb{E}_\eta[T(X)], \qquad \nabla^2 A(\eta) = \mathrm{Cov}_\eta(T(X)). \tag{0.3.4}$$

In particular A is convex. Strict convexity corresponds, after removing redundant components, to a nondegenerate natural statistic.

0.3.2 Minimal Sufficiency

Sufficient statistics are not unique: if T is sufficient, then (T, U) is sufficient for any statistic U. We therefore seek the coarsest sufficient reduction.

Definition 0.3.2 (Minimal sufficient statistic)**.** A sufficient statistic T is *minimal sufficient* if, for every sufficient statistic S, there exists a measurable function q such that

$$T = q(S) \qquad P_\theta\text{-almost surely for every } \theta.$$

Equivalently, the σ-field generated by T is contained, up to common null sets, in that generated by every other sufficient statistic.

Minimality does not mean that T has the smallest Euclidean dimension. A one-to-one encoding can map a vector into a scalar without changing its information. The relevant object is the partition of the sample space into level sets of T, or equivalently the σ-field $\sigma(T)$.

Theorem 0.3.2 (Likelihood-ratio criterion)**.** *Suppose the model is dominated and its densities have common support $\mathcal{X}_0$. If a statistic T satisfies*

$$T(x) = T(y) \quad \Longleftrightarrow \quad \frac{f_\theta(x)}{f_\theta(y)} \text{ is constant as a function of } \theta, \tag{0.3.5}$$

for $x, y \in \mathcal{X}_0$, then T is minimal sufficient. With suitable measure-theoretic qualifications, the criterion remains valid beyond the strict common-support setting.

Idea of proof. The forward implication in (0.3.5) yields a factorization and hence sufficiency. If S is any sufficient statistic, then $S(x) = S(y)$ forces the likelihood ratio $f_\theta(x)/f_\theta(y)$ to be free of θ. The reverse implication in (0.3.5) then forces $T(x) = T(y)$. Thus every level set of S lies inside a level set of T, so T is a function of S. □

Example 0.3.6 (Minimal sufficiency for coin flips)**.** For two strings $x, y \in \{0,1\}^n$ and $0 < p < 1$,

$$\frac{f_p(x)}{f_p(y)} = \left(\frac{p}{1-p}\right)^{\sum_i x_i - \sum_i y_i}.$$

This ratio is constant in p if and only if $\sum_i x_i = \sum_i y_i$. Therefore the number of heads is minimal sufficient for p.

Parameter-dependent support requires care but often makes the criterion especially informative.

Example 0.3.7 (Uniform endpoint)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$. The joint density is

$$f_\theta(x) = \theta^{-n} \mathbb{1}\{0 \le x_{(1)},\ x_{(n)} \le \theta\}.$$

Because the lower endpoint is known to be zero, this factors through the sample maximum $X_{(n)}$. For samples x, y in the nonnegative orthant, the likelihood ratio is independent of θ exactly when $x_{(n)} = y_{(n)}$; if the maxima differ, there are values of θ for which one likelihood is zero and the other is not. Hence $X_{(n)}$ is minimal sufficient.

0.3.3 Complete Sufficiency

Sufficiency says that a statistic retains all information in the sample. Completeness says that its distribution contains enough variation across θ to distinguish integrable functions of that statistic.

Definition 0.3.3 (Completeness)**.** A statistic T is *complete* for $\mathcal{P}$ if, for every measurable function a for which $\mathbb{E}_\theta\,|a(T)| < \infty$ for all θ,

$$\mathbb{E}_\theta[a(T)] = 0 \quad \text{for every } \theta \in \Theta \qquad \Longrightarrow \qquad a(T) = 0 \quad P_\theta\text{-almost surely for every } \theta.$$

It is *boundedly complete* if the implication is required only for bounded a.

Completeness is a property of a family of distributions of T, not of a single distribution. It will later guarantee uniqueness of unbiased estimators that are functions of a sufficient statistic.

Theorem 0.3.3 (Completeness of full exponential families)**.** *Consider the exponential family* (0.3.3)*. If its natural parameter space contains a nonempty open subset of* $\mathbb{R}^k$*, then the natural statistic* T *is complete, subject to the stated integrability condition.*

Proof sketch. Suppose $\mathbb{E}_\eta[a(T)] = 0$ throughout an open set. After multiplying by $e^{A(\eta)}$, this identity says that the Laplace transform of a signed measure induced by $a(T)h(X)\mu(\,\mathrm{d}x)$ vanishes on an open set. Uniqueness of Laplace transforms implies that this signed measure is zero, and therefore $a(T) = 0$ almost surely throughout the family. □

The openness condition matters. If a k-dimensional exponential family is restricted to a lower-dimensional curve, a nonzero function may have mean zero at every parameter value in the restricted family.

Example 0.3.8 (Completeness of the binomial count)**.** Let $S \sim \text{Binomial}(n, p)$, $0 < p < 1$, and suppose $\mathbb{E}_p[a(S)] = 0$ for every p. Then

$$0 = \sum_{s=0}^{n} a(s) \binom{n}{s} p^s (1-p)^{n-s}.$$

Dividing by $(1-p)^n$ and putting $z = p/(1-p) > 0$ gives

$$\sum_{s=0}^{n} a(s) \binom{n}{s} z^s = 0 \qquad \text{for every } z > 0.$$

A polynomial that vanishes on an interval has every coefficient equal to zero, so $a(s) = 0$ for $s = 0, \ldots, n$. Thus S is complete. Together with the preceding results, S is complete and sufficient for p.

Example 0.3.9 (Sufficient but not complete)**.** Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(\theta - 1, \theta + 1)$. The pair $T = (X_{(1)}, X_{(n)})$ is sufficient for θ. The range $R = X_{(n)} - X_{(1)}$ is a nonconstant function of T, but its distribution does not depend on θ. Hence

$$a(T) = R - \mathbb{E}_\theta[R]$$

has expectation zero for every θ without being almost surely zero. Therefore T is not complete.

Proposition 0.3.1. *If T is complete and sufficient, then T is minimal sufficient (up to common null sets).*

Proof sketch. Let S be any sufficient statistic and let $B \in \sigma(T)$. By sufficiency of S, a version of $q(S) = \mathbb{E}_\theta[\mathbb{1}_B \mid S]$ can be chosen independent of θ. By sufficiency of T and iterated conditioning, $\mathbb{E}_\theta[q(S) \mid T]$ is also parameter-free. Its difference from $\mathbb{1}_B$ has mean zero for every θ; completeness forces that difference to vanish. This ultimately shows that B is measurable with respect to S modulo common null sets. Hence $\sigma(T) \subseteq \sigma(S)$. ☐

0.3.4 Rao-Blackwell Theorem

Sufficiency is not only a statement about compression; it gives a systematic way to improve statistical procedures. Given an estimator $U = U(X)$ and a sufficient statistic T, define its Rao–Blackwellization by

$$U^*(T) = \mathbb{E}_\theta[U \mid T].$$

Because T is sufficient, the conditional distribution of X given T is parameter-free. Consequently the right-hand side can be chosen as a single function of T that does not involve the unknown θ.

Theorem 0.3.4 (Rao–Blackwell)**.** *Let T be sufficient for θ, let U be an estimator of $g(\theta)$, and suppose the relevant expectations exist. If $a \mapsto L(\theta, a)$ is convex for every θ, then*

$$R(\theta, U^*) \leq R(\theta, U) \qquad \text{for every } \theta. \tag{0.3.6}$$

If the loss is strictly convex, equality at a given θ can hold only when $U = U^*$ almost surely under P_θ (apart from degeneracies in the loss).

Proof. Conditional Jensen's inequality gives

$$L(\theta, \mathbb{E}_\theta[U \mid T]) \leq \mathbb{E}_\theta[L(\theta, U) \mid T].$$

Taking P_θ-expectations proves (0.3.6). Sufficiency ensures that U^* is a statistic rather than a rule involving the unknown parameter. The equality statement follows from the equality condition in strict Jensen's inequality. ☐

Under squared-error loss, the result has an exact variance decomposition. If U has finite second moment, then

$$\mathrm{Var}_\theta(U) = \mathrm{Var}_\theta(\mathbb{E}_\theta[U \mid T]) + \mathbb{E}_\theta[\mathrm{Var}_\theta(U \mid T)]\,. \tag{0.3.7}$$

Moreover $\mathbb{E}_\theta[U^*] = \mathbb{E}_\theta[U]$, so Rao–Blackwellization preserves bias. Its mean squared error can only decrease, and the reduction equals $\mathbb{E}_\theta[\mathrm{Var}_\theta(U \mid T)]$.

Example 0.3.10 (One flip versus all flips)**.** Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. The estimator $U = X_1$ is unbiased for p, and $S = \sum_i X_i$ is sufficient. Given $S = s$, each of the n positions is equally likely to contain any one of the s heads, so

$$U^* = \mathbb{E}_p[X_1 \mid S] = \frac{S}{n} = \bar{X}.$$

The variances are

$$\mathrm{Var}_p(X_1) = p(1-p), \qquad \mathrm{Var}_p(\bar{X}) = \frac{p(1-p)}{n}.$$

Thus using the sufficient statistic reduces variance by a factor of n.

Example 0.3.11 (Estimating the probability of all heads)**.** In the same model, $U = \prod_{i=1}^{n} X_i$ is unbiased for p^n. Since $U = 1$ exactly when $S = n$, it is already a function of the sufficient statistic and Rao–Blackwellization makes no change. By contrast, for $m \le n$, $U_m = \prod_{i=1}^{m} X_i$ is unbiased for p^m and

$$\mathbb{E}_p[U_m \mid S = s] = \frac{\binom{s}{m}}{\binom{n}{m}},$$

where the ratio is defined as zero for $s < m$. This is the conditional probability that a fixed set of m positions is contained among the s heads, and it has no dependence on p.

Rao–Blackwellization need not produce a unique or globally optimal estimator by itself. Completeness supplies the missing uniqueness.

Theorem 0.3.5 (Lehmann–Scheffé)**.** *If T is complete and sufficient and U is any unbiased estimator of $g(\theta)$ with finite variance, then*

$$\delta(T) = \mathbb{E}_\theta[U \mid T]$$

is the unique unbiased estimator of $g(\theta)$ that is a function of T, and it has variance no larger than that of any unbiased estimator. Thus it is the uniformly minimum variance unbiased estimator of $g(\theta)$.

Proof. Rao–Blackwellization shows that $\delta(T)$ is unbiased and has no larger variance than U. If $\delta_1(T)$ and $\delta_2(T)$ are two unbiased functions of T, then their difference has expectation zero for all θ. Completeness implies $\delta_1(T) = \delta_2(T)$ almost surely. Applying Rao–Blackwellization to any competing unbiased estimator proves the minimum-variance claim. $\square$

For the coin model, the binomial count S is complete and sufficient. Consequently S/n is the unique uniformly minimum variance unbiased estimator of p, and $\binom{S}{m}/\binom{n}{m}$ is the corresponding estimator of p^m. These examples summarize the main data-reduction pipeline:

$$\text{raw sample} \longrightarrow \text{sufficient statistic} \longrightarrow \text{Rao–Blackwellized rule},$$

with completeness converting the improved rule into a uniqueness and optimality statement.

Part II

Optimal Estimation, Exact

0.4 Uniformly Minimum Variance Unbiased Estimators

Let $g : \Theta \to \mathbb{R}$ be the estimand. An estimator $\delta(X)$ is *unbiased* for $g(\theta)$ if

$$\mathbb{E}_\theta[\delta(X)] = g(\theta) \qquad \text{for every } \theta \in \Theta.$$

Under squared-error loss, its risk decomposes as

$$R(\theta, \delta) = \mathbb{E}_\theta[(\delta - g(\theta))^2] = \mathrm{Var}_\theta(\delta) + \{\mathbb{E}_\theta[\delta] - g(\theta)\}^2. \tag{0.4.1}$$

Thus risk equals variance within the class of unbiased estimators.

Definition 0.4.1 (UMVU estimator)**.** An unbiased estimator δ_0 of $g(\theta)$ is a *uniformly minimum variance unbiased* (UMVU) estimator if

$$\mathrm{Var}_\theta(\delta_0) \le \mathrm{Var}_\theta(\delta)$$

for every θ and every unbiased estimator δ of $g(\theta)$. The word “uniformly” means that the same estimator minimizes variance at every parameter value.

Pointwise minimization is not automatic: an estimator that has smaller variance near one parameter value may have larger variance elsewhere. A useful characterization turns the minimization problem into an orthogonality condition.

Theorem 0.4.1 (Orthogonality characterization)**.** *Suppose δ_0 is unbiased for $g(\theta)$ and has finite second moment. Then δ_0 is UMVU if and only if*

$$\mathrm{Cov}_\theta(\delta_0, U) = 0 \tag{0.4.2}$$

for every finite-variance estimator U satisfying $\mathbb{E}_\theta[U] = 0$ for all θ.

Proof. For every real a, $\delta_0 + aU$ is unbiased and

$$\mathrm{Var}_\theta(\delta_0 + aU) = \mathrm{Var}_\theta(\delta_0) + 2a\mathrm{Cov}_\theta(\delta_0, U) + a^2\mathrm{Var}_\theta(U).$$

If δ_0 is UMVU, this quadratic is minimized at $a = 0$, forcing the covariance to vanish. Conversely, the difference $U = \delta - \delta_0$ between any two unbiased estimators has mean zero. Orthogonality then gives $\mathrm{Var}_\theta(\delta) = \mathrm{Var}_\theta(\delta_0) + \mathrm{Var}_\theta(U)$. □

The main construction is the Lehmann–Scheffé theorem from the preceding part: find a complete sufficient statistic T, begin with any unbiased estimator, and condition on T.

Example 0.4.1 (Powers of a coin bias)**.** Let $S \sim \mathrm{Binomial}(n, p)$. Since S is complete and sufficient, the UMVU estimator of p is S/n. More generally, for $1 \le m \le n$,

$$\widehat{p^m} = \frac{(S)_m}{(n)_m} = \frac{\binom{S}{m}}{\binom{n}{m}},$$

where $(a)_m = a(a-1)\cdots(a-m+1)$. Indeed, $\mathbb{E}_p[(S)_m] = (n)_m p^m$. No unbiased estimator of p^m exists when $m > n$: the expectation of any function of S is a polynomial in p of degree at most n.

Example 0.4.2 (Normal mean and variance)**.** If $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, then $(\sum_i X_i, \sum_i X_i^2)$ is complete and sufficient because the normal family is a full-rank exponential family with an open natural parameter space. Therefore $\bar{X}$ is UMVU for μ, and

$$S_X^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

is UMVU for σ^2.

0.5 Uniformly Minimum Risk Unbiased Estimators

Variance is tied specifically to scalar estimation under squared-error loss. For a general loss, an analogous restricted optimum is useful.

Definition 0.5.1 (UMRU estimator)**.** Within a specified class $\mathcal{U}$ of unbiased decision rules, a rule δ_0 is *uniformly minimum risk unbiased* (UMRU) if

$$R(\theta, \delta_0) \le R(\theta, \delta) \qquad \text{for every } \theta \in \Theta, \quad \delta \in \mathcal{U}.$$

For scalar estimation, $\mathcal{U}$ usually consists of rules satisfying $\mathbb{E}_\theta[\delta] = g(\theta)$.

Under squared-error loss, UMRU and UMVU are the same notion by (0.4.1). Under another loss they can differ. For example, unbiasedness constrains a mean, whereas absolute-error risk is governed by the entire distribution and is naturally minimized by medians.

There is also an intrinsic definition of unbiasedness for a decision problem that does not require a vector-valued action to have a mean.

Definition 0.5.2 (Loss unbiasedness)**.** A decision rule δ is *risk unbiased*, or *unbiased in loss*, if

$$\mathbb{E}_\theta[L(\theta, \delta(X))] \le \mathbb{E}_\theta[L(\theta', \delta(X))] \quad \text{for all } \theta, \theta' \in \Theta. \tag{0.5.1}$$

Thus, when the data come from P_θ, the rule is on average at least as close to the true target θ as to any false target θ'.

For $L(\theta, a) = (a - \theta)^2$ and finite second moments, subtracting the two sides of (0.5.1) shows that it holds for every θ' if and only if $\mathbb{E}_\theta[\delta] = \theta$. Hence ordinary and risk unbiasedness agree in this important case.

If T is sufficient and the loss is convex in the action, conditioning a rule on T preserves ordinary unbiasedness and reduces risk by conditional Jensen. Consequently the search for a UMRU rule may be restricted to functions of T. Completeness then provides uniqueness when the relevant unbiasedness equations reduce to expectations of functions of T.

0.6 Information Inequality

The information inequality supplies a lower bound on the variance of any unbiased estimator without explicitly comparing it with all competitors. Assume first that θ is scalar and that

P_θ has density f_θ with respect to a fixed measure. The *score* and Fisher information are

$$\ell'_\theta(X) = \frac{\partial}{\partial\theta}\log f_\theta(X), \qquad I_X(\theta) = \mathbb{E}_\theta[(\ell'_\theta(X))^2].$$

Under regularity conditions allowing differentiation under the integral and requiring parameter-independent support,

$$\mathbb{E}_\theta[\ell'_\theta(X)] = 0, \qquad I_X(\theta) = -\mathbb{E}_\theta[\ell''_\theta(X)].$$

Theorem 0.6.1 (Cramér–Rao information inequality)**.** *Suppose δ has finite variance, $\mathbb{E}_\theta[\delta] = g(\theta)$, and the usual differentiation and support conditions hold. If $0 < I_X(\theta) < \infty$, then*

$$\mathrm{Var}_\theta(\delta) \geq \frac{\{g'(\theta)\}^2}{I_X(\theta)}. \tag{0.6.1}$$

Equality holds at θ precisely when $\delta - g(\theta)$ is almost surely proportional to the score.

Proof. Differentiating $\mathbb{E}_\theta[\delta] = g(\theta)$ gives

$$g'(\theta) = \mathbb{E}_\theta[\delta\ell'_\theta] = \mathrm{Cov}_\theta(\delta, \ell'_\theta).$$

Cauchy–Schwarz implies $\{g'(\theta)\}^2 \leq \mathrm{Var}_\theta(\delta)I_X(\theta)$. Its equality condition gives the final assertion. □

For iid observations, scores add and have mean zero, so information adds: $I_{X_1,\ldots,X_n}(\theta) = nI_{X_1}(\theta)$.

Example 0.6.1 (Weighted coin)**.** For one Bernoulli(p) observation,

$$\ell'_p(X) = \frac{X-p}{p(1-p)}, \qquad I_X(p) = \frac{1}{p(1-p)}.$$

For n flips, every unbiased estimator of p has variance at least $p(1-p)/n$. The sample proportion S/n attains the bound because

$$\frac{S}{n} - p = \frac{p(1-p)}{n}\ell'_p(X_1, \ldots, X_n).$$

It is therefore efficient and UMVU.

If $\theta \in \mathbb{R}^k$, define the score vector and information matrix by

$$s_\theta(X) = \nabla_\theta \log f_\theta(X), \qquad I_X(\theta) = \mathbb{E}_\theta[s_\theta s_\theta^\top].$$

For an unbiased estimator of a scalar $g(\theta)$, the matrix form is

$$\mathrm{Var}_\theta(\delta) \geq \nabla g(\theta)^\top I_X(\theta)^{-1}\nabla g(\theta). \tag{0.6.2}$$

For a vector estimator D with mean $g(\theta)$, the corresponding covariance matrix inequality is

$$\mathrm{Cov}_\theta(D) \succeq \mathrm{D}g(\theta) I_X(\theta)^{-1} \mathrm{D}g(\theta)^\mathsf{T},$$

where $A \succeq B$ means that $A - B$ is positive semidefinite.

The regularity assumptions cannot be ignored. For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{Uniform}(0, \theta)$, the support depends on θ, and differentiating the integral as above misses a boundary term. Indeed, a rescaled sample maximum is unbiased and can have variance smaller than a naive application of the regular Cramér–Rao formula would permit.

0.7 Minimum Risk Equivariant Estimators

Many models possess symmetries. Let a group G act on the sample space, parameter space, and action space. A problem is *invariant* if transforming both the data and parameter leaves the model and loss unchanged:

$$X \sim P_\theta \Longrightarrow gX \sim P_{g\theta}, \qquad L(g\theta, ga) = L(\theta, a).$$

A rule δ is *equivariant* if

$$\delta(gx) = g\delta(x).$$

For an equivariant rule, risk is constant along group orbits: $R(g\theta, \delta) = R(\theta, \delta)$. When the group acts transitively on Θ, its risk is constant over the whole parameter space. A *minimum risk equivariant* (MRE) rule minimizes this constant among all equivariant rules.

0.7.1 Location and Scale Families

In a location family, $X_i = \mu + Z_i$ for a known distribution of the errors. The translation group sends x to $x + c\mathbf{1}$ and μ to $\mu + c$. Under loss $L(\mu, a) = \rho(a - \mu)$, an equivariant estimator satisfies

$$\delta(x + c\mathbf{1}) = \delta(x) + c.$$

Its error distribution does not depend on μ, and hence its risk is constant.

Under squared-error loss, the MRE location estimator is the generalized Bayes rule under the translation-invariant prior $\pi(\mu) \propto 1$:

$$\delta_{\mathrm{MRE}}(x) = \frac{\int_{-\infty}^{\infty} \mu f(x - \mu\mathbf{1})\,\mathrm{d}\mu}{\int_{-\infty}^{\infty} f(x - \mu\mathbf{1})\,\mathrm{d}\mu}, \tag{0.7.1}$$

provided these integrals exist. This is often called the Pitman estimator. For normal errors with known variance it reduces to $\bar{X}$.

In a positive scale family, $X_i = \sigma Z_i$, $\sigma > 0$. The scale group sends x to cx, σ to $c\sigma$, and an equivariant estimator of σ obeys $\delta(cx) = c\delta(x)$. A scale-invariant loss, such as

$$L(\sigma, a) = \left(\frac{a}{\sigma} - 1\right)^2,$$

makes the risk of every scale-equivariant rule constant. The invariant measure on the scale group leads to the improper prior $\pi(\sigma) \propto 1/\sigma$; the resulting generalized Bayes rule is a candidate MRE estimator.

Improper priors do not define probability distributions, so generalized Bayes calculations require separate justification. The posterior integrals must be finite, and optimality among equivariant rules does not by itself imply optimality among all rules.

0.7.2 Risk Unbiasedness

Equivariance and unbiasedness impose different restrictions. Equivariance requires a rule to respect the geometry of the problem; risk unbiasedness requires the true parameter to minimize expected loss as in (0.5.1). Neither condition generally implies the other.

For a location problem under squared-error loss, every equivariant rule can be written schematically as

$$\delta(X) = X_1 + h(X_2 - X_1, \ldots, X_n - X_1).$$

Its bias is constant in μ. It is risk unbiased exactly when that constant is zero. Thus risk unbiasedness may select a proper subset of the equivariant class, while MRE optimality selects the rule with smallest constant risk in the larger class.

The distinction matters in constrained parameter spaces. If a parameter is known to lie in an interval, an estimator can often be improved under squared error by projecting it onto that interval. Projection generally introduces bias but decreases loss pointwise. Hence restricting attention to unbiased rules can exclude sensible procedures.

0.8 Average Risk Optimality

A risk function gives one value for every θ, and two risk functions may cross. One way to obtain a scalar criterion is to average risk with respect to a probability distribution π on Θ. The *Bayes risk* is

$$r(\pi, \delta) = \int_\Theta R(\theta, \delta)\, \pi(\,\mathrm{d}\theta). \tag{0.8.1}$$

This does not require a subjective interpretation: π can also be viewed as a device for weighting regions of the parameter space.

0.8.1 Bayes Estimators

By Tonelli's theorem, the Bayes risk can be written using the posterior:

$$r(\pi, \delta) = \int_{\mathcal{X}} \left\{ \int_\Theta L(\theta, \delta(x)) \pi(\,\mathrm{d}\theta \mid x) \right\} m_\pi(\,\mathrm{d}x),$$

where m_π is the prior predictive distribution. Therefore a Bayes rule may be found by minimizing posterior expected loss separately for each x.

Proposition 0.8.1 (Bayes actions)**.** *For a scalar target $g(\theta)$, a Bayes estimator is*

- *the posterior mean under squared-error loss;*
- *any posterior median under absolute-error loss;*
- *a posterior mode under zero-one loss in a discrete action space.*

Example 0.8.1 (Beta–binomial estimation)**.** Let $S \mid p \sim \text{Binomial}(n,p)$ and $p \sim \text{Beta}(\alpha,\beta)$, where $\alpha,\beta > 0$. Conjugacy gives

$$p \mid S = s \sim \text{Beta}(\alpha+s, \beta+n-s).$$

Under squared-error loss the Bayes estimator is

$$\delta_\pi(s) = \mathbb{E}[p \mid S=s] = \frac{\alpha+s}{\alpha+\beta+n} = w\frac{s}{n} + (1-w)\frac{\alpha}{\alpha+\beta}, \tag{0.8.2}$$

where $w = n/(n+\alpha+\beta)$. It shrinks the observed proportion toward the prior mean. Its posterior squared-error loss is minimized uniquely, and its Bayes risk equals the expected posterior variance.

Theorem 0.8.1 (Admissibility of a unique Bayes rule)**.** *If a Bayes rule δ_π has finite Bayes risk, is unique up to prior predictive null sets, and every nonempty set on which a competing rule could strictly improve has positive prior mass, then δ_π is admissible. For example, the last condition follows when π has full support and the risk difference is continuous in θ.*

Proof. If a rule dominated δ_π, integrating the risk inequality against π would give no larger Bayes risk. Strict improvement on a set of positive prior mass would give strictly smaller Bayes risk, contradicting Bayes optimality. Uniqueness rules out domination that differs only on sets invisible to the prior predictive distribution. □

0.9 Worst Case Optimality

When no averaging distribution is accepted, a conservative criterion is the maximum risk

$$\overline{R}(\delta) = \sup_{\theta\in\Theta} R(\theta,\delta).$$

This treats nature as selecting the least favorable parameter after the rule has been chosen.

0.9.1 Minimax Estimators

Definition 0.9.1 (Minimax rule)**.** A rule δ^* is minimax if

$$\sup_\theta R(\theta,\delta^*) = \inf_\delta \sup_\theta R(\theta,\delta).$$

The common value is the minimax value of the decision problem.

For every prior π and rule δ,

$$\inf_{\tilde{\delta}} \sup_{\theta} R(\theta, \tilde{\delta}) \geq \inf_{\tilde{\delta}} r(\pi, \tilde{\delta}) = r(\pi, \delta_\pi). \tag{0.9.1}$$

Thus every Bayes risk is a lower bound on the minimax value.

Theorem 0.9.1 (Constant-risk Bayes criterion)**.** *If a Bayes rule δ_π has constant risk $R(\theta, \delta_\pi) = c$ for all θ in the parameter space, then it is minimax and its Bayes risk is c.*

Proof. The maximum risk of δ_π is c. On the other hand, (0.9.1) says that the minimax value is at least its Bayes risk, also c. □

The theorem explains why invariant rules with constant risk are natural minimax candidates. A generalized Bayes rule from an improper invariant prior cannot be inserted directly into the proof, but it is often obtained as a limit of proper Bayes rules whose Bayes risks approach its constant risk.

0.9.2 Least Favorable Priors

Definition 0.9.2 (Least favorable prior)**.** A prior π^* is *least favorable* if it maximizes the optimal Bayes risk:

$$r(\pi^*, \delta_{\pi^*}) = \sup_{\pi} \inf_{\delta} r(\pi, \delta).$$

It concentrates weight where the estimation problem is hardest.

If a prior π^* and its Bayes rule satisfy

$$\sup_{\theta} R(\theta, \delta_{\pi^*}) = r(\pi^*, \delta_{\pi^*}),$$

then the lower and upper bounds meet: δ_{π^*} is minimax and π^* is least favorable. In finite games this is the familiar saddle point condition.

Often no proper least favorable prior exists. It is enough to find priors π_m whose Bayes risks converge to the maximum risk of a candidate rule: if

$$r(\pi_m, \delta_{\pi_m}) \longrightarrow c \quad \text{and} \quad \sup_{\theta} R(\theta, \delta^*) = c,$$

then (0.9.1) proves that δ^* is minimax. Such a sequence is called a least favorable sequence.

Example 0.9.1 (Normal location)**.** If $X \sim \mathcal{N}(\theta, \sigma^2)$ and loss is squared error, the estimator $\delta(X) = X$ has constant risk σ^2. Take proper priors $\theta \sim \mathcal{N}(0, \tau^2)$. The Bayes rule is

$$\delta_\tau(X) = \frac{\tau^2}{\tau^2 + \sigma^2} X,$$

and its Bayes risk is $\sigma^2\tau^2/(\sigma^2 + \tau^2) \uparrow \sigma^2$ as $\tau^2 \to \infty$. Hence these priors form a least favorable sequence and X is minimax.

0.10 Simultaneous Estimation

Suppose $X = (X_1, \dots, X_d)^\mathsf{T} \sim \mathcal{N}_d(\theta, I_d)$ and the goal is to estimate the entire mean vector under total squared-error loss

$$L(\theta, a) = \|a - \theta\|^2 .$$

The coordinatewise estimator $\delta_0(X) = X$ is unbiased, equivariant, and has constant risk

$$R(\theta, \delta_0) = \mathbb{E}_\theta \|X - \theta\|^2 = d.$$

It is minimax, since it is the limiting Bayes rule under increasingly diffuse normal priors. Surprisingly, minimaxity does not imply admissibility.

0.10.1 James-Stein Estimator

For $d \geq 3$, define

$$\delta_{\mathrm{JS}}(X) = \left(1 - \frac{d-2}{\|X\|^2}\right) X. \tag{0.10.1}$$

This estimator shrinks X toward the origin. Although every coordinate of X is individually the usual estimator of its mean, estimating the coordinates jointly permits an improvement in total risk.

Lemma 0.10.1 (Stein's identity)**.** *If $X \sim \mathcal{N}_d(\theta, I_d)$ and $g : \mathbb{R}^d \to \mathbb{R}^d$ is sufficiently regular and integrable, then*

$$\mathbb{E}_\theta[(X - \theta)^\mathsf{T} g(X)] = \mathbb{E}_\theta[\operatorname{div} g(X)].$$

Proof. Apply one-dimensional integration by parts to each coordinate and sum. The normal density satisfies $\partial f_\theta(x)/\partial x_i = -(x_i - \theta_i) f_\theta(x)$, and the boundary terms vanish under the integrability assumptions. □

For an estimator $X + g(X)$, Stein's identity gives the unbiased risk formula

$$R(\theta, X + g) - d = \mathbb{E}_\theta \left[\|g(X)\|^2 + 2 \operatorname{div} g(X)\right]. \tag{0.10.2}$$

Set $g(x) = -ax/\|x\|^2$. Away from the origin,

$$\|g(x)\|^2 = \frac{a^2}{\|x\|^2}, \qquad \operatorname{div} g(x) = -\frac{a(d-2)}{\|x\|^2}.$$

Therefore

$$R(\theta, \delta_a) - d = \{a^2 - 2a(d-2)\} \mathbb{E}_\theta \left[\frac{1}{\|X\|^2}\right].$$

For $d \geq 3$ this expectation is finite, and every $0 < a < 2(d-2)$ strictly improves on X for all θ. The choice $a = d - 2$ gives (0.10.1) and maximizes the displayed risk reduction.

The positive-part version

$$\delta^+_{\mathrm{JS}}(X) = \left(1 - \frac{d-2}{\|X\|^2}\right)_+ X$$

dominates the ordinary James–Stein rule. Shrinkage toward another fixed point m follows by replacing X with $X - m$ and adding m back. The choice of center expresses structural information and can have a substantial effect on realized error, even though the dominance theorem holds for every fixed center.

This phenomenon does not contradict the scalar Cramér–Rao bound: the James–Stein estimator is biased, and performance is measured by the sum of squared errors across coordinates. It also shows why admissibility, UMVU optimality, and minimaxity are distinct concepts.

0.11 Empirical Bayes Estimators

Hierarchical models provide a principled source of shrinkage. Suppose $\theta_1, \dots, \theta_m$ are drawn from a distribution G_λ with unknown hyperparameter λ, and conditionally independent observations satisfy $X_i \mid \theta_i \sim P_{\theta_i}$. If λ were known, the Bayes rule would be

$$\delta_\lambda(x_i) = \mathbb{E}_\lambda[\theta_i \mid X_i = x_i]$$

under squared-error loss. An *empirical Bayes* rule estimates λ from the ensemble $X_1, \dots, X_m$ and plugs the estimate into the Bayes rule.

Example 0.11.1 (Many weighted coins)**.** For coin i, observe $S_i \mid p_i \sim \text{Binomial}(n_i, p_i)$ and assume $p_i \overset{\text{iid}}{\sim} \text{Beta}(\alpha, \beta)$. If α and β were known, then

$$\mathbb{E}[p_i \mid S_i] = \frac{\alpha + S_i}{\alpha + \beta + n_i}.$$

Estimate (α, β) from the marginal beta-binomial likelihood of all counts, or by matching the across-coin mean and variance. The empirical Bayes estimator is

$$\widehat{p}_i^{\text{EB}} = \frac{\widehat{\alpha} + S_i}{\widehat{\alpha} + \widehat{\beta} + n_i}.$$

Coins with few flips are shrunk strongly toward the estimated population mean $\widehat{\alpha}/(\widehat{\alpha} + \widehat{\beta})$; coins with many flips are driven mainly by their own sample proportions.

Example 0.11.2 (Normal means)**.** Suppose $X_i \mid \theta_i \sim \mathcal{N}(\theta_i, \sigma^2)$ and $\theta_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \tau^2)$, with σ^2 known. The oracle Bayes estimator is

$$\mathbb{E}[\theta_i \mid X_i] = \mu + B(X_i - \mu), \qquad B = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

Replacing μ and τ^2 by estimates from the marginal model $X_i \sim \mathcal{N}(\mu, \tau^2 + \sigma^2)$ yields an empirical Bayes shrinkage rule. A simple moment estimate is

$$\widehat{\mu} = \bar{X}, \qquad \widehat{\tau}^2 = \max\left\{0, \frac{1}{m-1}\sum_{i=1}^{m}(X_i - \bar{X})^2 - \sigma^2\right\}.$$

Empirical Bayes procedures borrow strength across related problems, but their analysis must account for estimation of the prior. Plug-in posterior variances generally understate uncertainty because they treat $\widehat{\lambda}$ as fixed. Parametric bootstrap, asymptotic corrections, or a fully Bayesian hyperprior can propagate this additional uncertainty. Moreover, shrinkage is only as appropriate as the exchangeability assumption: if the units come from substantially different populations, pooling can introduce systematic bias. These considerations separate the construction of an empirical Bayes point estimator from the calibration of its uncertainty statements.

Part III

Optimal Hypothesis Testing, Exact

0.12 Introduction to Hypothesis Testing

A hypothesis divides the parameter space into a null set Θ_0 and an alternative set Θ_1. We write

$$H_0 : \theta \in \Theta_0 \qquad \text{versus} \qquad H_1 : \theta \in \Theta_1.$$

A nonrandomized test is a measurable map $\phi : \mathcal{X} \to \{0, 1\}$, where $\phi(X) = 1$ means that H_0 is rejected. It is convenient to allow randomized tests $\phi : \mathcal{X} \to [0, 1]$, interpreted as the conditional probability of rejection after observing $X = x$.

The *power function* of a test is

$$\beta_\phi(\theta) = \mathbb{E}_\theta[\phi(X)]. \tag{0.12.1}$$

For $\theta \in \Theta_0$, this is the probability of a type I error; for $\theta \in \Theta_1$, $1 - \beta_\phi(\theta)$ is the probability of a type II error. The *size* is

$$\sup_{\theta \in \Theta_0} \beta_\phi(\theta),$$

and a test has *level* α if its size is at most α. Size equals level only when the supremum is attained at α.

Testing is a decision problem with actions 0 and 1. If false rejection has loss c_0 and false acceptance has loss c_1, then

$$R(\theta, \phi) = \begin{cases} c_0 \beta_\phi(\theta), & \theta \in \Theta_0, \\ c_1 \{1 - \beta_\phi(\theta)\}, & \theta \in \Theta_1. \end{cases}$$

The frequentist level-α formulation fixes an upper bound on the first kind of risk and then seeks large power on the alternative.

Example 0.12.1 (Testing whether a coin is fair)**.** For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$, consider $H_0 : p = 1/2$ against $H_1 : p > 1/2$. Large values of $S = \sum_i X_i$ constitute evidence against H_0. A test of the form

$$\phi(s) = \mathbb{1}\{s > c\} + \gamma \mathbb{1}\{s = c\}$$

can attain any desired level α: choose c so that $P_{1/2}(S > c) \le \alpha \le P_{1/2}(S \ge c)$ and set

$$\gamma = \frac{\alpha - P_{1/2}(S > c)}{P_{1/2}(S = c)}.$$

Randomization is needed only because the binomial distribution is discrete. In practice, one often uses the conservative nonrandomized test with $\gamma = 0$.

A p-value is a statistic measuring how extreme the observation is relative to a null distribution. For a test statistic T where large values are evidence against H_0, a valid p-value $P(X)$ satisfies

$$\sup_{\theta \in \Theta_0} P_\theta\{P(X) \le u\} \le u, \qquad 0 \le u \le 1.$$

Rejecting when $P(X) \le \alpha$ therefore gives a level-α test. A p-value is not the posterior probability that the null is true.

0.13 Simple Hypotheses

Both hypotheses are *simple* when they specify individual distributions:

$$H_0 : \theta = \theta_0 \qquad \text{versus} \qquad H_1 : \theta = \theta_1.$$

This case is completely solved by ordering observations according to their likelihood ratio.

0.13.1 Most Powerful (MP) Tests

Definition 0.13.1 (Most powerful test)**.** A level-α test ϕ^* is *most powerful* for testing P_{θ_0} against P_{θ_1} if

$$\mathbb{E}_{\theta_1}[\phi^*] \geq \mathbb{E}_{\theta_1}[\phi]$$

for every other level-α test ϕ.

The level constraint is essential. Without it, the rule that always rejects has power one but provides no control of false rejection. Randomization also makes the feasible class convex and permits exact size in discrete models.

Example 0.13.1 (One weighted-coin flip)**.** Test $H_0 : p = p_0$ against $H_1 : p = p_1$, where $p_1 > p_0$, based on a single flip. Heads has likelihood ratio p_1/p_0, whereas tails has ratio $(1 - p_1)/(1 - p_0)$; the former is larger. For $\alpha \leq p_0$, the MP test rejects with probability α/p_0 after heads and never after tails. For $\alpha > p_0$, it always rejects after heads and rejects after tails with probability $(\alpha - p_0)/(1 - p_0)$.

0.13.2 Neyman-Pearson Lemma

Theorem 0.13.1 (Neyman–Pearson lemma)**.** *Suppose P_0 and P_1 have densities f_0 and f_1 with respect to a common dominating measure. Choose $k \geq 0$ and $\gamma \in [0, 1]$ so that*

$$\phi^*(x) = \begin{cases} 1, & f_1(x) > kf_0(x), \\ \gamma, & f_1(x) = kf_0(x), \\ 0, & f_1(x) < kf_0(x) \end{cases} \quad \textit{satisfies} \quad \mathbb{E}_0[\phi^*] = \alpha. \tag{0.13.1}$$

Then ϕ^* is most powerful among all tests of level α. Conversely, every MP test of size α has this form, apart from null sets and possible freedom on the boundary, provided the two densities are handled on their zero sets in the usual way.

Proof. For any level-α test ϕ,

$$\int (\phi^* - \phi)(f_1 - kf_0)\, \mathrm{d}\mu \geq 0,$$

because both factors have the same sign pointwise. Hence

$$\mathbb{E}_1[\phi^*] - \mathbb{E}_1[\phi] \geq k\{\mathbb{E}_0[\phi^*] - \mathbb{E}_0[\phi]\} \geq 0.$$

This proves sufficiency. The equality conditions in these inequalities yield the converse characterization. □

Example 0.13.2 (Several coin flips)**.** For $S \sim \text{Binomial}(n, p)$ and $p_1 > p_0$,

$$\frac{f_{p_1}(s)}{f_{p_0}(s)} = \left(\frac{1-p_1}{1-p_0}\right)^n \left\{\frac{p_1(1-p_0)}{p_0(1-p_1)}\right\}^s,$$

which is strictly increasing in s. Therefore an MP test rejects for large numbers of heads, with possible randomization at one boundary count.

0.13.3 Likelihood Ratio Tests

For a general null and alternative, define the generalized likelihood ratio statistic

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} f_\theta(x)}{\sup_{\theta \in \Theta} f_\theta(x)}, \qquad 0 \le \Lambda(x) \le 1. \tag{0.13.2}$$

The *likelihood ratio test* (LRT) rejects for small Λ. For two simple hypotheses this is equivalent to the Neyman–Pearson likelihood-ratio ordering. For composite hypotheses, however, the Neyman–Pearson lemma does not automatically make the LRT optimal.

Example 0.13.3 (Two-sided normal mean)**.** If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known, consider $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$. The unrestricted MLE is $\widehat{\mu} = \bar{X}$, and direct calculation gives

$$-2 \log \Lambda = \frac{n(\bar{X} - \mu_0)^2}{\sigma^2}.$$

Thus the level-α LRT rejects when $\sqrt{n}\left|\bar{X} - \mu_0\right|/\sigma > z_{1-\alpha/2}$. Its finite-sample null distribution is exactly standard normal after taking the signed square root; no asymptotic approximation is needed.

Example 0.13.4 (Two-sided coin test)**.** For $S \sim \text{Binomial}(n, p)$ and $H_0 : p = p_0$ against $p \neq p_0$, the unrestricted MLE is $\widehat{p} = S/n$. The LRT statistic is

$$\Lambda(s) = \frac{p_0^s(1-p_0)^{n-s}}{(s/n)^s(1-s/n)^{n-s}},$$

with $0^0 = 1$. Rejection occurs in both tails, but discreteness means the two tail probabilities generally cannot be balanced exactly without randomization.

0.14 One-Sided Hypotheses

Now consider an ordered scalar parameter and hypotheses such as

$$H_0 : \theta \le \theta_0 \qquad \text{versus} \qquad H_1 : \theta > \theta_0.$$

For many families, larger values of a statistic T systematically favor larger values of θ. This ordering can turn the collection of simple-alternative Neyman–Pearson tests into a single optimal test.

0.14.1 Uniformly Most Powerful (UMP) Tests

Definition 0.14.1 (UMP test)**.** A level-α test ϕ^* is *uniformly most powerful* for $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ if

$$\beta_{\phi^*}(\theta) \geq \beta_\phi(\theta) \quad \text{for every } \theta \in \Theta_1$$

and every other level-α test ϕ.

The same test must be MP against every point in the alternative. Such a test often exists for one-sided problems because the likelihood-ratio ordering is the same for all alternatives on one side. It rarely exists for genuinely two-sided problems: an MP test against a smaller parameter rejects in the lower tail, while one against a larger parameter rejects in the upper tail.

For a boundary-null problem, the size is often attained at the boundary θ_0. This must be proved, usually by showing that the rejection probability is monotone in θ.

0.14.2 Monotone Likelihood Ratio (MLR)

Definition 0.14.2 (Monotone likelihood ratio)**.** A family $\{f_\theta : \theta \in \Theta \subseteq \mathbb{R}\}$ has a *monotone likelihood ratio* (MLR) in $T(X)$ if, whenever $\theta_1 > \theta_0$, the ratio $f_{\theta_1}(x)/f_{\theta_0}(x)$ is a nondecreasing function of $T(x)$ on the common support.

Theorem 0.14.1 (Karlin–Rubin)**.** *Suppose the family has MLR in T. Choose c and γ so that*

$$\phi^*(x) = \mathbb{1}\{T(x) > c\} + \gamma \mathbb{1}\{T(x) = c\}$$

has rejection probability α at θ_0. Then ϕ^* is UMP level α for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$, provided its rejection probability is nondecreasing in θ. Reversing the tails gives the analogous result for a lower-sided alternative.

Proof sketch. For every fixed $\theta_1 > \theta_0$, the Neyman–Pearson lemma and MLR imply that the same upper-tail test is MP for θ_0 versus θ_1. Monotonicity of its power ensures level α throughout the composite null. Since it is MP at every point in the alternative, it is UMP. $\square$

Example 0.14.1 (One-sided weighted-coin test)**.** The binomial family has MLR in S because the likelihood ratio displayed above is increasing in s whenever $p_1 > p_0$. Consequently, the test that rejects for large S is UMP level α for $H_0 : p \leq p_0$ against $H_1 : p > p_0$. Its exact power function is

$$\beta(p) = P_p(S > c) + \gamma P_p(S = c),$$

which is nondecreasing in p by stochastic ordering of binomial variables.

A regular one-parameter exponential family

$$f_\eta(x) = h(x) \exp\{\eta T(x) - A(\eta)\}$$

has MLR in T, since for $\eta_1 > \eta_0$ the likelihood ratio is a positive constant times $e^{(\eta_1 - \eta_0)T(x)}$. Thus the theorem covers many classical one-sided tests at once.

0.15 Composite Hypotheses

When both Θ_0 and Θ_1 contain multiple distributions, no single ordering of the sample points need be optimal. Useful approaches include finding a UMP test when special structure permits it, restricting the class of tests (for example to unbiased or invariant tests), using a likelihood ratio test, or optimizing average or worst-case power.

The role of sufficiency persists. If T is sufficient and $\phi(X)$ is any test, then

$$\phi^*(T) = \mathbb{E}_\theta[\phi(X) \mid T]$$

is a well-defined, parameter-free randomized test with exactly the same power function as ϕ. Thus attention may be restricted to tests based on a sufficient statistic without loss.

0.15.1 Uniformly Most Powerful (UMP) Tests

UMP tests for composite alternatives exist only when the MP rejection regions can be made compatible across all alternative points. The one-sided MLR setting is the principal example. Two-sided alternatives usually require a weaker optimality criterion.

Proposition 0.15.1 (Nonexistence in a two-sided normal problem)**.** *Let $X \sim \mathcal{N}(\mu, 1)$ and test $H_0 : \mu = 0$ against $H_1 : \mu \neq 0$ at a nontrivial level α. No UMP test exists.*

Proof. By the Neyman–Pearson lemma, an MP test against any fixed $\mu_1 > 0$ rejects for large X, whereas an MP test against any fixed $\mu_1 < 0$ rejects for small X. Apart from trivial levels, no test can have both forms. □

One response is to impose invariance under $x \mapsto -x$, leading to a two-tailed test based on $|X|$. Another is to seek a UMP unbiased test. Both restrictions rule out tests that spend nearly all type I error in one tail.

0.15.2 Least Favorable Priors

A test can be derived as a Bayes rule by placing priors π_0 and π_1 on the null and alternative. The resulting mixture densities are

$$m_j(x) = \int_{\Theta_j} f_\theta(x)\pi_j(\,\mathrm{d}\theta), \qquad j = 0, 1.$$

For fixed prior odds and error costs, the Bayes test compares $m_1(x)/m_0(x)$ with a constant. Thus the Neyman–Pearson lemma applies to the two mixture distributions even though the original hypotheses are composite.

A least favorable prior places mass on parameter values for which control of error or attainment of power is hardest. If a Bayes test against the induced mixtures has its null rejection probability bounded by α everywhere and its worst alternative power equal to its average power under π_1, then it attains a testing saddle point. The Bayes lower bound and the worst-case upper bound coincide, proving maximin optimality.

For a one-sided MLR problem, the point mass at the boundary θ_0 is often least favorable for size because the rejection probability is largest there over the null. With nuisance parameters or separated composite hypotheses, least favorable priors can be nondegenerate mixtures and may only exist as limiting sequences.

0.16 Unbiased Tests

The power of a level-α test can fall below α at some alternative values. Such a test rejects less often under those alternatives than under the null boundary, an undesirable behavior that motivates a restriction.

Definition 0.16.1 (Unbiased test)**.** A level-α test ϕ for $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$ is *unbiased* if

$$\beta_\phi(\theta_1) \geq \beta_\phi(\theta_0) \quad \text{for every } \theta_1 \in \Theta_1,\ \theta_0 \in \Theta_0.$$

In the common exact-size formulation this becomes $\beta_\phi(\theta) \leq \alpha$ on Θ_0 and $\beta_\phi(\theta) \geq \alpha$ on Θ_1.

A test is *uniformly most powerful unbiased* (UMPU) if it maximizes power at every alternative point among all level-α unbiased tests.

Consider a regular one-parameter exponential family and the two-sided problem $H_0 : \eta = \eta_0$ against $H_1 : \eta \neq \eta_0$. If the power function is differentiable, unbiasedness makes η_0 a local minimum, so

$$\beta_\phi(\eta_0) = \alpha, \qquad \beta'_\phi(\eta_0) = 0. \tag{0.16.1}$$

Differentiating under the integral and using the exponential-family score gives

$$\beta'_\phi(\eta_0) = \mathbb{E}_{\eta_0}[\phi(X)\{T - \mathbb{E}_{\eta_0}T\}].$$

Thus the constraints in (0.16.1) are equivalent to

$$\mathbb{E}_{\eta_0}[\phi] = \alpha, \qquad \mathbb{E}_{\eta_0}[T\phi] = \alpha\mathbb{E}_{\eta_0}[T].$$

The UMPU test rejects for $T < c_1$ or $T > c_2$, with possible boundary randomization, where the two constants are chosen to satisfy these size and moment constraints. This allocates rejection probability between both tails in the manner required by unbiasedness.

Example 0.16.1 (Two-sided normal mean)**.** For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known σ, the UMPU level-α test of $H_0 : \mu = \mu_0$ against $\mu \neq \mu_0$ rejects when

$$\frac{\sqrt{n}\left|\bar{X} - \mu_0\right|}{\sigma} > z_{1-\alpha/2}.$$

Symmetry supplies the moment constraint, and the power is at least α for every $\mu \neq \mu_0$.

0.16.1 Unbiased Tests with Nuisance Parameters

Suppose the parameter is (ψ, λ), where ψ is of interest and λ is a nuisance parameter. A level-α test must control

$$\sup_{\lambda} \mathbb{E}_{\psi_0,\lambda}[\phi(X)],$$

which can be difficult if the null distribution depends on λ.

A test is *similar* if

$$\mathbb{E}_{\psi_0,\lambda}[\phi(X)] = \alpha \qquad \text{for every } \lambda. \tag{0.16.2}$$

Similarity is stronger than level control and is closely connected to unbiasedness for two-sided problems. Conditioning often removes the nuisance parameter exactly.

Suppose that under H_0 a statistic U is sufficient and complete for λ. If ϕ is similar, then

$$\mathbb{E}_{\psi_0,\lambda}\left[\mathbb{E}_{\psi_0,\lambda}(\phi \mid U) - \alpha\right] = 0 \quad \text{for every } \lambda.$$

Completeness implies

$$\mathbb{E}_{\psi_0,\lambda}[\phi(X) \mid U] = \alpha \quad \text{almost surely.} \tag{0.16.3}$$

The conditional distribution used to construct the test is free of λ, converting the problem into conditional simple-null tests.

Example 0.16.2 (Normal mean with unknown variance)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, with σ^2 unknown. The statistic

$$T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{S_X}$$

has a t_{n-1} distribution under H_0, independent of σ^2. Therefore

$$\phi(X) = \mathbb{1}\{|T| > t_{n-1,1-\alpha/2}\}$$

is similar of level α. In the normal exponential family it is also the UMP unbiased test for the two-sided mean problem.

Example 0.16.3 (Conditional test for two binomial samples)**.** Let $S_j \sim \text{Binomial}(n_j, p_j)$ independently and test $H_0 : p_1 = p_2 = p$ against unequal success probabilities. Under H_0, the total $S_1 + S_2$ is sufficient for the nuisance parameter p, and

$$P(S_1 = s_1 \mid S_1 + S_2 = s) = \frac{\binom{n_1}{s_1}\binom{n_2}{s-s_1}}{\binom{n_1+n_2}{s}},$$

a hypergeometric probability independent of p. Tail probabilities from this conditional distribution yield Fisher's exact test. Boundary randomization can make it exactly similar; without randomization it is typically conservative.

Conditioning is powerful but changes the criterion: optimality is established within the similar or unbiased class, not necessarily among all level tests. This qualification is important whenever a procedure is described as "optimal" in the presence of nuisance parameters.

Part IV

Optimal Estimation, Asymptotic

0.17 Introduction to Asymptotic Estimation

Exact optimality compares estimators at a fixed sample size. Such comparisons can be difficult or overly restrictive: an unbiased estimator may not exist, the finite-sample distribution may be intractable, and several procedures may differ only by terms that vanish as the sample grows. Asymptotic theory studies a sequence of experiments and estimators

$$\delta_n = \delta_n(X_1, \dots, X_n)$$

as $n \to \infty$.

Three questions organize the theory:

1. Does δ_n approach the target? This is consistency.
2. At what rate does its error vanish, and what is the limiting distribution of the rescaled error?
3. Can another regular estimator have a smaller limiting risk or covariance?

For regular finite-dimensional iid models, the canonical rate is $n^{-1/2}$ and the canonical limit is Gaussian. Neither conclusion is automatic. Boundary parameters, nonidentification, parameter-dependent support, heavy tails, and nonsmooth criteria may produce different rates or non-Gaussian limits.

Example 0.17.1 (Weighted coin)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ and $\widehat{p}_n = \bar{X}_n$. The law of large numbers gives $\widehat{p}_n \xrightarrow{p} p$, while the central limit theorem gives, for $0 < p < 1$,

$$\sqrt{n}(\widehat{p}_n - p) \xrightarrow{d} \mathcal{N}(0, p(1-p)).$$

Consistency describes the eventual location of the estimator; asymptotic normality describes the local shape and scale of its errors.

0.18 Review

All limits below are taken under P_θ unless another distribution is specified. Since the probability measure changes with θ, asymptotic claims should state whether they hold pointwise for each fixed θ or uniformly over a subset of the parameter space.

0.18.1 Convergence

Definition 0.18.1 (Modes of stochastic convergence)**.** For random variables Y_n and Y:

- $Y_n \xrightarrow{a.s.} Y$ if $P(\lim_n Y_n = Y) = 1$;
- $Y_n \xrightarrow{p} Y$ if, for every $\varepsilon > 0$, $P(|Y_n - Y| > \varepsilon) \to 0$;
- $Y_n \xrightarrow{d} Y$ if $\mathbb{E}[f(Y_n)] \to \mathbb{E}[f(Y)]$ for every bounded continuous f;

- $Y_n \to Y$ in L^r if $\mathbb{E}\,|Y_n - Y|^r \to 0$.

The definitions extend to random vectors by replacing absolute value with a norm.

The usual implications are

$$L^r \Longrightarrow \text{probability} \Longrightarrow \text{distribution}, \qquad \text{almost surely} \Longrightarrow \text{probability}.$$

Convergence in distribution to a constant implies convergence in probability to that constant. In general, convergence in distribution does not imply convergence of moments; uniform integrability is needed to control tails.

Theorem 0.18.1 (Continuous mapping and Slutsky)**.** *If $Y_n \xrightarrow{d} Y$ and g is continuous at Y almost surely, then $g(Y_n) \xrightarrow{d} g(Y)$. If additionally $Z_n \xrightarrow{p} c$, then*

$$Y_n + Z_n \xrightarrow{d} Y + c, \qquad Y_n Z_n \xrightarrow{d} cY,$$

and $Y_n/Z_n \xrightarrow{d} Y/c$ when $c \neq 0$.

Stochastic order notation summarizes rates. We write $Y_n = O_p(a_n)$ if Y_n/a_n is bounded in probability, and $Y_n = o_p(a_n)$ if $Y_n/a_n \xrightarrow{p} 0$. Thus

$$\sqrt{n}(\delta_n - \theta) = O_p(1) \quad \Longleftrightarrow \quad \delta_n - \theta = O_p(n^{-1/2}).$$

Theorem 0.18.2 (Weak law and central limit theorem)**.** *If $X_1, X_2, \ldots$ are iid with mean μ and finite variance σ^2, then*

$$\bar{X}_n \xrightarrow{p} \mu, \qquad \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2).$$

If only $\mathbb{E}\,|X|_1 < \infty$, the first conclusion still holds.

0.18.2 Consistency

Definition 0.18.2 (Consistency)**.** An estimator δ_n of $g(\theta)$ is *weakly consistent* at θ if $\delta_n \xrightarrow{p} g(\theta)$ under P_θ. It is strongly consistent if $\delta_n \xrightarrow{a.s.} g(\theta)$. It is uniformly consistent on $K \subseteq \Theta$ if, for every $\varepsilon > 0$,

$$\sup_{\theta \in K} P_\theta\{\|\delta_n - g(\theta)\| > \varepsilon\} \longrightarrow 0.$$

Consistency alone gives neither a useful uncertainty scale nor good finite-sample performance. If δ_n is consistent, so is $\delta_n + n^{-1/4} Z$ for many bounded random perturbations, even though the perturbed estimator converges much more slowly than a root-n estimator.

Two common proof strategies are direct moment bounds and optimization of a sample criterion. If $\mathbb{E}_\theta \delta_n \to g(\theta)$ and $\mathrm{Var}_\theta(\delta_n) \to 0$, Chebyshev's inequality proves consistency. For an M-estimator

$$\widehat{\theta}_n \in \arg\max_{\vartheta \in \Theta} M_n(\vartheta),$$

one instead shows that M_n converges uniformly to a deterministic function M uniquely maximized at θ.

Theorem 0.18.3 (Argmax consistency)**.** *Suppose* Θ *is compact,*

$$\sup_{\vartheta\in\Theta} |M_n(\vartheta) - M(\vartheta)| \xrightarrow{p} 0,$$

and M *is uniquely and well-separatedly maximized at* θ_0*: for every neighborhood* U *of* θ_0*,*

$$M(\theta_0) > \sup_{\vartheta\notin U} M(\vartheta).$$

If $M_n(\widehat{\theta}_n) \geq \sup_\vartheta M_n(\vartheta) - o_p(1)$*, then* $\widehat{\theta}_n \xrightarrow{p} \theta_0$*.*

Proof sketch. Outside any neighborhood U, the population criterion has a fixed gap below its maximum. Uniform convergence makes the sample criterion preserve this gap with probability approaching one. An approximate maximizer therefore cannot lie outside U with nonvanishing probability. □

0.18.3 Asymptotic Normality

Definition 0.18.3 (Asymptotic normality)**.** An estimator is asymptotically normal with rate $r_n \to \infty$, asymptotic bias $b(\theta)$, and asymptotic covariance $V(\theta)$ if

$$r_n\{\delta_n - g(\theta)\} \xrightarrow{d} \mathcal{N}\{b(\theta), V(\theta)\}.$$

In regular parametric models, $r_n = \sqrt{n}$.

The most useful derivations start from an asymptotically linear expansion

$$\delta_n - g(\theta) = \frac{1}{n}\sum_{i=1}^{n} \psi_\theta(X_i) + o_p(n^{-1/2}), \tag{0.18.1}$$

where $\mathbb{E}_\theta[\psi_\theta(X)] = 0$ and $\mathbb{E}_\theta \|\psi_\theta(X)\|^2 < \infty$. The function ψ_θ is the influence function. The central limit theorem and Slutsky's theorem give

$$\sqrt{n}\{\delta_n - g(\theta)\} \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}_\theta[\psi_\theta\psi_\theta^\mathsf{T}]\right).$$

An estimated standard error $\widehat{V}_n^{1/2}/\sqrt{n}$ is useful only if $\widehat{V}_n \xrightarrow{p} V(\theta)$. Studentization then gives

$$\frac{\sqrt{n}\{\delta_n - g(\theta)\}}{\sqrt{\widehat{V}_n}} \xrightarrow{d} \mathcal{N}(0,1)$$

in the scalar case, producing asymptotic confidence intervals.

0.18.4 Delta Method

Theorem 0.18.4 (Delta method)**.** *If*

$$r_n(T_n - \theta) \xrightarrow{d} Z$$

and $g : \mathbb{R}^k \to \mathbb{R}^m$ is differentiable at θ, then

$$r_n\{g(T_n) - g(\theta)\} \xrightarrow{d} \mathrm{D}g(\theta)Z. \tag{0.18.2}$$

In particular, if $Z \sim \mathcal{N}_k(0, V)$, the limiting covariance is $\mathrm{D}g(\theta)V\mathrm{D}g(\theta)^\mathsf{T}$.

Proof. Differentiability gives

$$g(T_n) - g(\theta) = \mathrm{D}g(\theta)(T_n - \theta) + o(\|T_n - \theta\|).$$

Consistency and $r_n(T_n - \theta) = O_p(1)$ make the rescaled remainder $o_p(1)$. Slutsky's theorem proves the result. □

Example 0.18.1 (Log odds of a coin)**.** For $0 < p < 1$, let $g(p) = \log\{p/(1-p)\}$. Since $g'(p) = 1/\{p(1-p)\}$ and $\sqrt{n}(\widehat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1-p))$,

$$\sqrt{n}\left[\log \frac{\widehat{p}}{1-\widehat{p}} - \log \frac{p}{1-p}\right] \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{p(1-p)}\right).$$

This approximation is an interior result; the log odds is undefined when the sample contains all heads or all tails and becomes unstable near $p = 0$ or $p = 1$.

If the first derivative vanishes, the first-order delta method is degenerate. If $g'(\theta) = 0$ but $g''(\theta) \neq 0$ and $\sqrt{n}(T_n - \theta) \xrightarrow{d} Z$, a second-order expansion yields

$$n\{g(T_n) - g(\theta)\} \xrightarrow{d} \frac{1}{2}g''(\theta)Z^2.$$

0.19 Asymptotic Unbiasedness

There are several inequivalent definitions of asymptotic unbiasedness. A basic one requires

$$\mathbb{E}_\theta[\delta_n] - g(\theta) \longrightarrow 0. \tag{0.19.1}$$

Consistency does not by itself imply (0.19.1), because rare but very large errors may prevent convergence of expectations. Consistency plus uniform integrability of $\{\delta_n\}$ does imply it.

At the root-n scale, one may instead require

$$\sqrt{n}\{\mathbb{E}_\theta[\delta_n] - g(\theta)\} \longrightarrow 0.$$

This stronger condition rules out first-order asymptotic bias. If

$$\sqrt{n}\{\delta_n - g(\theta)\} \xrightarrow{d} \mathcal{N}\{b(\theta), V(\theta)\}$$

and the rescaled sequence is uniformly integrable, then its limiting scaled bias is $b(\theta)$.

The distinction matters for mean squared error. If moment convergence is valid, then

$$n\mathbb{E}_\theta[(\delta_n - g(\theta))^2] \longrightarrow V(\theta) + b(\theta)^2. \tag{0.19.2}$$

A small bias can reduce variance enough to improve total risk, just as in finite-sample shrinkage. Asymptotic unbiasedness is therefore a restriction, not a universal requirement of good estimation.

Example 0.19.1 (Smoothed coin estimator)**.** For constants $a, b > 0$, the beta-prior posterior mean is

$$\widetilde{p}_n = \frac{S + a}{n + a + b}.$$

Its bias is

$$\mathbb{E}_p[\widetilde{p}_n] - p = \frac{a - (a+b)p}{n + a + b} = O(n^{-1}),$$

so it is unbiased to first order on the root-n scale and has the same asymptotic distribution as S/n. Nevertheless, its finite-sample risk and boundary behavior differ from those of the sample proportion.

0.20 Asymptotic Efficiency

If two consistent estimators have root-n limits

$$\sqrt{n}(\delta_{j,n} - g(\theta)) \xrightarrow{d} \mathcal{N}(0, V_j(\theta)), \qquad j = 1, 2,$$

then estimator 1 is asymptotically at least as efficient as estimator 2 when $V_2(\theta) - V_1(\theta)$ is positive semidefinite. In the scalar case, the asymptotic relative efficiency is often written

$$\mathrm{ARE}(1, 2) = \frac{V_2(\theta)}{V_1(\theta)}.$$

It approximates the factor by which estimator 2 needs a larger sample to match estimator 1's variance.

Pointwise asymptotic variance alone is too weak for a general optimality theory. An estimator can be modified at a single parameter value to converge faster there while behaving normally elsewhere.

Example 0.20.1 (Hodges' estimator)**.** Let $\bar{X}_n \sim \mathcal{N}(\theta, 1/n)$ and define

$$\delta_n = \begin{cases} 0, & \left|\bar{X}_n\right| \le n^{-1/4}, \\ \bar{X}_n, & \left|\bar{X}_n\right| > n^{-1/4}. \end{cases}$$

At $\theta = 0$, $\sqrt{n}\delta_n \xrightarrow{p} 0$, apparently improving on the usual $\mathcal{N}(0, 1)$ limit. At every fixed $\theta \neq 0$, it has the same asymptotic distribution as $\bar{X}_n$. But along parameters approaching zero near the threshold, its risk is poor. This *superefficiency* shows why efficiency bounds require regularity under local alternatives, not merely a pointwise limit at each fixed parameter.

Definition 0.20.1 (Regular estimator)**.** In a locally asymptotically normal model, an estimator δ_n of a scalar or vector parameter is regular at θ if, for every fixed local direction h, the distribution of

$$\sqrt{n}\{\delta_n - (\theta + h/\sqrt{n})\}$$

under $P_{\theta+h/\sqrt{n}}$ converges to a limit that does not depend on h.

Regularity excludes the unstable local behavior of Hodges' estimator and is the appropriate class for asymptotic information bounds.

0.21 Asymptotic Information Inequality

Let $s_\theta(X) = \nabla_\theta \log f_\theta(X)$ and let $I(\theta) = \mathbb{E}_\theta[s_\theta s_\theta^\mathsf{T}]$. For a regular estimator of $g(\theta) \in \mathbb{R}^m$, the asymptotic analogue of the Cramér–Rao bound is

$$V(\theta) \succeq \mathrm{D}g(\theta)I(\theta)^{-1}\mathrm{D}g(\theta)^\mathsf{T}. \tag{0.21.1}$$

Unlike the finite-sample bound, this result allows small finite-sample bias; regularity supplies the local stability that replaces exact unbiasedness.

Theorem 0.21.1 (Hájek–Le Cam convolution theorem)**.** *In a regular LAN model, the limiting distribution of a regular estimator of θ has the representation*

$$Z = Z_0 + W, \qquad Z_0 \sim \mathcal{N}_k(0, I(\theta)^{-1}),$$

where Z_0 and W are independent. Consequently $\mathrm{Cov}(Z) \succeq I(\theta)^{-1}$ *whenever the covariance exists.*

The Gaussian component Z_0 is unavoidable sampling noise; W is additional noise contributed by an inefficient procedure. An estimator is *asymptotically efficient* if $W = 0$, so that

$$\sqrt{n}(\delta_n - \theta) \xrightarrow{d} \mathcal{N}_k(0, I(\theta)^{-1}).$$

The corresponding local asymptotic minimax theorem provides a risk statement that is robust to moving parameters. For suitable bowl-shaped loss L and every fixed radius c,

$$\liminf_{n\to\infty} \sup_{\|h\|\le c} \mathbb{E}_{\theta+h/\sqrt{n}} L\Big(\sqrt{n}\{\delta_n - (\theta + h/\sqrt{n})\}\Big)$$

is bounded below, as $c \to \infty$, by the risk of $Z_0 \sim \mathcal{N}_k(0, I(\theta)^{-1})$. This formulation rules out gains at a single point purchased by poor behavior in a shrinking neighborhood.

Example 0.21.1 (Efficiency for the coin bias)**.** One Bernoulli observation has information $I(p) = 1/\{p(1-p)\}$. Hence the asymptotic variance lower bound for regular estimators of p is $I(p)^{-1} = p(1-p)$. Since

$$\sqrt{n}(\widehat{p} - p) \xrightarrow{d} \mathcal{N}(0, p(1-p)),$$

the sample proportion is asymptotically efficient at every interior $p \in (0,1)$. The regular theory does not apply at $p = 0$ or $p = 1$.

0.22 Maximum Likelihood Estimators

For a dominated model, an MLE is any measurable maximizer

$$\widehat{\theta}_n \in \arg\max_{\theta\in\Theta} \ell_n(\theta), \qquad \ell_n(\theta) = \sum_{i=1}^{n} \log f_\theta(X_i).$$

The MLE is invariant under one-to-one reparameterization: if $\widehat{\theta}_n$ maximizes the likelihood and g is one-to-one, then $g(\widehat{\theta}_n)$ is the MLE of $g(\theta)$. For non-one-to-one g, the same conclusion holds when the likelihood for $g(\theta)$ is defined by profiling over its fibers.

0.22.1 Consistency and Inconsistency

If the data come from P_{θ_0}, the normalized log-likelihood tends pointwise to

$$M(\theta) = \mathbb{E}_{\theta_0}[\log f_\theta(X)].$$

The difference from its value at the truth is

$$M(\theta) - M(\theta_0) = -\operatorname{KL}(P_{\theta_0} \| P_\theta) \leq 0. \tag{0.22.1}$$

Thus identifiability makes θ_0 the unique population maximizer. To transfer this fact to the sample maximizer, pointwise laws of large numbers are generally insufficient; one needs uniform control and must prevent the MLE from escaping to the boundary or infinity.

Theorem 0.22.1 (A consistency theorem for MLEs)**.** *Suppose Θ is compact, the model is identifiable, the map $\theta \mapsto \log f_\theta(x)$ is suitably continuous, and a uniform law of large numbers holds:*

$$\sup_{\theta\in\Theta} \left| n^{-1}\ell_n(\theta) - M(\theta) \right| \xrightarrow{p} 0.$$

If M is well-separatedly maximized at θ_0, then every approximate MLE is consistent for θ_0.

Example 0.22.1 (Coin MLE)**.** For $S \sim \operatorname{Binomial}(n, p)$, the log-likelihood is

$$\ell_n(p) = S \log p + (n - S) \log(1 - p),$$

and the MLE is $\widehat{p} = S/n$, including the boundary values when all flips agree. The law of large numbers immediately proves consistency.

MLEs may be inconsistent when the model is misspecified, the parameter is not identified, the dimension grows too quickly, or maximization exploits an unbounded likelihood.

Example 0.22.2 (Misspecification)**.** If the true distribution is P_0 but the fitted family $\{P_\theta\}$ does not contain it, the MLE generally converges to the pseudo-true parameter

$$\theta^* = \arg\min_\theta \operatorname{KL}(P_0 \| P_\theta),$$

not to a literal data-generating parameter. Inference then uses a sandwich covariance rather than the inverse Fisher information unless the model is correctly specified.

Example 0.22.3 (An unbounded likelihood)**.** In a normal mixture with an unknown component variance, a component mean can be placed on one observation while its variance tends to zero. The likelihood then diverges, so an unconstrained global MLE does not exist. Large sample size does not repair a nonexistent maximizer; constraints or penalization are required.

0.22.2 Asymptotic Normality and Efficiency

The score equation is $\nabla \ell_n(\widehat{\theta}_n) = 0$. For a scalar interior parameter, Taylor expansion around the truth gives

$$0 = \ell_n'(\theta_0) + \ell_n''(\widetilde{\theta}_n)(\widehat{\theta}_n - \theta_0),$$

where $\widetilde{\theta}_n$ lies between $\widehat{\theta}_n$ and θ_0. Therefore

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = \left\{-\frac{1}{n}\ell_n''(\widetilde{\theta}_n)\right\}^{-1} \frac{\ell_n'(\theta_0)}{\sqrt{n}}. \tag{0.22.2}$$

Under regularity and consistency, the first factor converges to $I(\theta_0)^{-1}$ and the second converges to $\mathcal{N}(0, I(\theta_0))$. Hence

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1}). \tag{0.22.3}$$

The MLE attains the asymptotic information bound and is efficient.

Equation (0.22.2) also gives the asymptotically linear representation

$$\widehat{\theta}_n - \theta_0 = \frac{1}{n}\sum_{i=1}^{n} I(\theta_0)^{-1} s_{\theta_0}(X_i) + o_p(n^{-1/2}).$$

Thus the efficient influence function is $I(\theta_0)^{-1} s_{\theta_0}(X)$.

Common variance estimators are the inverse expected information, inverse observed information, and the outer product of scores. Under correct specification they are asymptotically equivalent. Under misspecification, if

$$A = -\mathbb{E}_0[\nabla_\theta^2 \log f_{\theta^*}(X)], \qquad B = \mathbb{E}_0[s_{\theta^*}(X) s_{\theta^*}(X)^\top],$$

then the limiting covariance is the sandwich matrix

$$A^{-1}BA^{-1}, \tag{0.22.4}$$

which reduces to $I(\theta_0)^{-1}$ when the information identity $A = B$ holds.

0.22.3 MLE for Multi-parameter Families

Let $\theta \in \mathbb{R}^k$. Under the multivariate analogues of consistency, interiority, differentiability, nonsingular information, and uniform control of the Hessian,

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = I(\theta_0)^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} s_{\theta_0}(X_i) + o_p(1) \xrightarrow{d} \mathcal{N}_k(0, I(\theta_0)^{-1}). \tag{0.22.5}$$

For a differentiable target $g : \mathbb{R}^k \to \mathbb{R}^m$, the delta method gives

$$\sqrt{n}\{g(\widehat{\theta}_n) - g(\theta_0)\} \xrightarrow{d} \mathcal{N}_m\left(0, \mathrm{D}g(\theta_0) I(\theta_0)^{-1} \mathrm{D}g(\theta_0)^\top\right).$$

Example 0.22.4 (Mean and variance of a normal distribution)**.** If $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, parameterize by $\theta = (\mu, \sigma^2)$. The MLEs are

$$\widehat{\mu} = \bar{X}, \qquad \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

The per-observation information matrix is diagonal:

$$I(\mu, \sigma^2) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 1/(2\sigma^4) \end{pmatrix}.$$

Consequently,

$$\sqrt{n}\begin{pmatrix} \widehat{\mu} - \mu \\ \widehat{\sigma}^2 - \sigma^2 \end{pmatrix} \xrightarrow{d} \mathcal{N}_2\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{pmatrix}\right).$$

Although $\widehat{\sigma}^2$ is biased at finite n, its bias is $O(n^{-1})$ and disappears on the root-n scale.

If $\theta = (\psi, \lambda)$ contains a nuisance parameter, the asymptotic covariance for estimating ψ is the corresponding block of $I(\theta)^{-1}$, equivalently the inverse Schur complement

$$\left(I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi}\right)^{-1}.$$

This is the same efficient information that governs local asymptotic testing. Estimation and testing are therefore two views of the same local Gaussian geometry.

Finally, (0.22.5) is an interior regular-model result. At boundaries, under singular information, or when the number of parameters grows with n, the MLE may have a different rate and limit. Those cases require a fresh expansion rather than mechanical use of inverse Fisher information.

Part V

Optimal Hypothesis Testing, Asymptotic

Exact finite-sample optimality is available only in specially structured models. In regular problems, asymptotic theory replaces the experiment based on n observations by a limiting Gaussian experiment. This approximation does more than provide critical values: it identifies the scale of statistically distinguishable alternatives and yields upper bounds on local power.

0.23 Asymptotic Testing Framework

Let $X^{(n)}$ denote the data in the nth experiment, with distribution $P_{n,\theta}$, and let $\phi_n(X^{(n)}) \in [0,1]$ be a sequence of tests. The models need not literally be iid, although that is the principal example.

Definition 0.23.1 (Asymptotic level)**.** The sequence $\{\phi_n\}$ has *pointwise asymptotic level* α if

$$\limsup_{n\to\infty} \mathbb{E}_{n,\theta}[\phi_n] \le \alpha \qquad \text{for every } \theta \in \Theta_0.$$

It has *uniform asymptotic level* α if

$$\limsup_{n\to\infty} \sup_{\theta\in\Theta_0} \mathbb{E}_{n,\theta}[\phi_n] \le \alpha. \tag{0.23.1}$$

Uniform validity is stronger: the parameter value producing the largest rejection probability may vary with n. Pointwise convergence alone does not exclude severe size distortion along such a sequence.

Definition 0.23.2 (Consistency of a test)**.** A sequence of tests is *consistent* against a fixed $\theta \in \Theta_1$ if

$$\mathbb{E}_{n,\theta}[\phi_n] \longrightarrow 1.$$

It is uniformly consistent over a set $A \subseteq \Theta_1$ if the infimum of its power over A converges to one.

Consistency against fixed alternatives is often easy and does not distinguish good tests from mediocre ones. More informative comparisons use local alternatives that approach the null as the sample grows.

0.23.1 Fixed and Local Alternatives

In a regular iid model, estimators fluctuate on the $n^{-1/2}$ scale. Thus the canonical local alternatives to a point null θ_0 are

$$\theta_n = \theta_0 + \frac{h}{\sqrt{n}}, \qquad h \in \mathbb{R}^k \text{ fixed.} \tag{0.23.2}$$

Alternatives separated from θ_0 by much more than $n^{-1/2}$ are typically detected with probability tending to one. Those separated by much less are typically indistinguishable from the null. At the $n^{-1/2}$ scale, power has a nondegenerate limit strictly between size and one.

Example 0.23.1 (Local alternatives for a weighted coin)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ and test $H_0 : p = p_0$, where $0 < p_0 < 1$. Define

$$Z_n = \frac{S - np_0}{\sqrt{np_0(1-p_0)}}.$$

Under $p = p_0$, the central limit theorem gives $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. Under $p_n = p_0 + h/\sqrt{n}$,

$$Z_n = \frac{S - np_n}{\sqrt{np_n(1-p_n)}}\sqrt{\frac{p_n(1-p_n)}{p_0(1-p_0)}} + \frac{h}{\sqrt{p_0(1-p_0)}},$$

so

$$Z_n \xrightarrow{d} \mathcal{N}\left(\frac{h}{\sqrt{p_0(1-p_0)}}, 1\right). \tag{0.23.3}$$

The upper-tailed test $\mathbb{1}\{Z_n > z_{1-\alpha}\}$ therefore has limiting local power

$$1 - \Phi\left(z_{1-\alpha} - \frac{h}{\sqrt{p_0(1-p_0)}}\right).$$

0.23.2 Asymptotic Critical Values and p-Values

Suppose a statistic T_n has continuous limiting distribution F_0 under the null. If large values are evidence against H_0, then

$$\phi_n = \mathbb{1}\{T_n > F_0^{-1}(1-\alpha)\}$$

has asymptotic rejection probability α at parameter values for which $T_n \xrightarrow{d} F_0$. The corresponding asymptotic p-value is $1 - F_0(T_n)$.

This argument proves pointwise validity only. Uniform validity requires a uniform distributional approximation over the composite null. It can fail near boundaries, where information degenerates, or when nuisance parameters are weakly identified. When a finite-sample null distribution is available, there is no reason to replace it with an asymptotic one.

0.24 Local Asymptotic Normality

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} P_\theta$ with log-likelihood $\ell_n(\theta) = \sum_{i=1}^n \log f_\theta(X_i)$. Write

$$s_\theta(X) = \nabla_\theta \log f_\theta(X), \qquad I(\theta) = \mathbb{E}_\theta[s_\theta(X)s_\theta(X)^\mathsf{T}].$$

The normalized score at θ_0 is

$$\Delta_n = \frac{1}{\sqrt{n}}\sum_{i=1}^n s_{\theta_0}(X_i).$$

Under regularity conditions, the multivariate central limit theorem gives $\Delta_n \xrightarrow{d} \mathcal{N}_k(0, I(\theta_0))$ under P_{θ_0}.

Definition 0.24.1 (Local asymptotic normality)**.** The sequence of experiments is *locally asymptotically normal* (LAN) at θ_0 if, for every fixed $h \in \mathbb{R}^k$,

$$\log \frac{\mathrm{d}P_{n,\theta_0+h/\sqrt{n}}}{\mathrm{d}P_{n,\theta_0}} = h^\mathsf{T}\Delta_n - \frac{1}{2}h^\mathsf{T}I(\theta_0)h + o_{P_{\theta_0}}(1), \tag{0.24.1}$$

with $\Delta_n \xrightarrow{d} \mathcal{N}_k(0, I(\theta_0))$ under the null.

A second-order Taylor expansion of the log-likelihood suggests (0.24.1); the LAN statement additionally controls the remainder under local perturbations. The limiting log-likelihood ratio is exactly that of observing

$$Y \sim \mathcal{N}_k(I(\theta_0)h, I(\theta_0))$$

with $h = 0$ under the null. Hence regular local testing problems inherit the geometry of a Gaussian shift experiment.

0.24.1 Contiguity

Definition 0.24.2 (Contiguity)**.** A sequence Q_n is *contiguous* with respect to P_n, written $Q_n \lhd P_n$, if $P_n(A_n) \to 0$ implies $Q_n(A_n) \to 0$ for every sequence of events A_n. If contiguity holds in both directions, the sequences are mutually contiguous.

In a regular LAN model, $P_{n,\theta_0+h/\sqrt{n}}$ and P_{n,θ_0} are mutually contiguous. Thus an event that is asymptotically negligible under the null remains negligible under local alternatives. This is what allows stochastic expansions proved under the null to remain useful for local-power calculations.

Theorem 0.24.1 (Le Cam's third lemma, Gaussian form)**.** *Suppose under P_n,*

$$\left(T_n, \log\frac{\mathrm{d}Q_n}{\mathrm{d}P_n}\right) \xrightarrow{d} \mathcal{N}\left(\begin{pmatrix}\mu \\ -\sigma^2/2\end{pmatrix}, \begin{pmatrix}V & c \\ c^\mathsf{T} & \sigma^2\end{pmatrix}\right).$$

Then under Q_n, $T_n \xrightarrow{d} \mathcal{N}(\mu + c, V)$, with the evident vector interpretation.

The lemma says that exponential tilting by the local likelihood ratio shifts the limiting mean by the covariance with that likelihood ratio. Applied to LAN, it yields

$$\Delta_n \xrightarrow{d} \mathcal{N}_k(I(\theta_0)h, I(\theta_0)) \quad \text{under } \theta_0 + h/\sqrt{n}. \tag{0.24.2}$$

0.24.2 Asymptotic Power Envelope

For a scalar parameter, standardize the central sequence as $Z_n = I(\theta_0)^{-1/2}\Delta_n$. Under the null it converges to $\mathcal{N}(0,1)$; under $\theta_0 + h/\sqrt{n}$ it converges to $\mathcal{N}(h\sqrt{I(\theta_0)}, 1)$. The Neyman–Pearson lemma in the limiting Gaussian experiment gives the one-sided power envelope

$$\pi^*(h) = 1 - \Phi\{z_{1-\alpha} - h\sqrt{I(\theta_0)}\}, \qquad h > 0. \tag{0.24.3}$$

A test whose limiting local power equals this expression is locally asymptotically most powerful in the specified direction.

For two-sided testing, no UMP test exists even in the limiting Gaussian experiment. Restricting to asymptotically unbiased or symmetric tests gives the rejection rule $|Z_n| > z_{1-\alpha/2}$ and local power

$$P\left(\left|Z + h\sqrt{I(\theta_0)}\right| > z_{1-\alpha/2}\right), \qquad Z \sim \mathcal{N}(0,1).$$

0.25 Likelihood-Based Tests

Let $\widehat{\theta}$ be the unrestricted maximum likelihood estimator and $\widetilde{\theta}$ the MLE under H_0. Three classical tests measure the same local departure from the null in different ways:

- the likelihood-ratio test compares maximized likelihoods;
- the Wald test measures the distance from $\widehat{\theta}$ to the null using estimated standard errors;
- the score test measures the slope of the likelihood at the constrained estimate $\widetilde{\theta}$.

0.25.1 Likelihood-Ratio Tests and Wilks' Theorem

Define

$$W_n = 2\{\ell_n(\widehat{\theta}) - \ell_n(\widetilde{\theta})\} = -2\log\Lambda_n. \tag{0.25.1}$$

Theorem 0.25.1 (Wilks)**.** *Suppose the model is regular, the true parameter is an interior point of Θ_0, and Θ_0 is a smooth submanifold of codimension r. Under H_0,*

$$W_n \xrightarrow{d} \chi^2_r.$$

Thus the asymptotic level-α LRT rejects when $W_n > \chi^2_{r,1-\alpha}$.

The degrees of freedom equal the number of independent restrictions imposed by the null. A quadratic expansion of ℓ_n shows that W_n is asymptotically the squared length of the normalized score component orthogonal to the null tangent space.

Under local alternatives, the limit is noncentral chi-squared:

$$W_n \xrightarrow{d} \chi^2_r(\lambda), \tag{0.25.2}$$

where λ is the squared information distance from the local direction h to the tangent space of the null. This limit determines local power.

0.25.2 Wald Tests

For a scalar restriction $H_0 : g(\theta) = 0$, the Wald statistic is

$$W_n^{\mathrm{Wald}} = \frac{g(\widehat{\theta})^2}{\nabla g(\widehat{\theta})^\top \{nI(\widehat{\theta})\}^{-1} \nabla g(\widehat{\theta})}. \tag{0.25.3}$$

Under regularity and the null, it converges to χ^2_1. For r smooth restrictions, replace the denominator by the corresponding covariance matrix and obtain a χ^2_r limit.

Wald tests are easy to construct from an estimator and its standard error, but their finite-sample behavior is not invariant to nonlinear reparameterization of the restriction. They can also behave poorly when the MLE is near a boundary or its estimated variance is unstable.

0.25.3 Score Tests

The score, or Lagrange multiplier, test needs only the null fit. For a scalar parameter and a simple null, its statistic is

$$W_n^{\text{score}} = \frac{\{\ell_n'(\theta_0)\}^2}{nI(\theta_0)}, \tag{0.25.4}$$

which converges to χ^2_1 under H_0. A signed one-sided version uses $\ell_n'(\theta_0)/\sqrt{nI(\theta_0)}$ directly.

Example 0.25.1 (Three asymptotic coin tests)**.** For $S \sim \text{Binomial}(n, p)$ and $H_0 : p = p_0$, the signed score statistic is

$$Z_{\text{score}} = \frac{S - np_0}{\sqrt{np_0(1 - p_0)}}.$$

The Wald statistic instead standardizes $\widehat{p} - p_0$ using $\widehat{p}(1 - \widehat{p})/n$, where $\widehat{p} = S/n$. The likelihood-ratio statistic is

$$2\left[S \log \frac{\widehat{p}}{p_0} + (n - S) \log \frac{1 - \widehat{p}}{1 - p_0}\right].$$

For fixed interior p_0, all three have the same first-order null limit and the same local asymptotic power. They can differ noticeably for small n or when p_0 is close to zero or one.

0.25.4 First-Order Equivalence

In a regular model with a scalar parameter,

$$2\{\ell_n(\widehat{\theta}) - \ell_n(\theta_0)\} = nI(\theta_0)(\widehat{\theta} - \theta_0)^2 + o_p(1) = \frac{\ell_n'(\theta_0)^2}{nI(\theta_0)} + o_p(1).$$

Consequently the likelihood-ratio, Wald, and score tests are first-order equivalent under the null and contiguous alternatives. This equivalence does not imply equal finite-sample performance or equal higher-order accuracy.

0.26 Nuisance Parameters and Efficient Scores

Partition the parameter as $\theta = (\psi, \lambda)$, where $\psi \in \mathbb{R}^r$ is the parameter of interest and λ is nuisance. Write the score and information in blocks:

$$s_\theta = \begin{pmatrix} s_\psi \\ s_\lambda \end{pmatrix}, \qquad I(\theta) = \begin{pmatrix} I_{\psi\psi} & I_{\psi\lambda} \\ I_{\lambda\psi} & I_{\lambda\lambda} \end{pmatrix}.$$

0.26.1 Efficient Score and Information

The component of s_ψ linearly predictable from the nuisance score does not provide information that distinguishes a change in ψ from a change in λ. Removing it gives the efficient score

$$s_{\psi\cdot\lambda} = s_\psi - I_{\psi\lambda} I_{\lambda\lambda}^{-1} s_\lambda. \tag{0.26.1}$$

Its covariance is the efficient information

$$I_{\psi\cdot\lambda} = I_{\psi\psi} - I_{\psi\lambda} I_{\lambda\lambda}^{-1} I_{\lambda\psi}, \tag{0.26.2}$$

the Schur complement of $I_{\lambda\lambda}$. Score tests with nuisance parameters use the efficient score evaluated at the restricted estimator.

Geometrically, nuisance scores span a tangent space in $L^2(P_\theta)$. Equation (0.26.1) projects the score for the parameter of interest onto the orthogonal complement of that space. Local power bounds are obtained by replacing ordinary information with efficient information.

0.26.2 Plug-In and Profile Likelihood

The profile log-likelihood is

$$\ell_n^p(\psi) = \sup_\lambda \ell_n(\psi, \lambda).$$

Comparing its constrained and unconstrained maxima yields the usual likelihood-ratio statistic. Under regularity, estimating the nuisance parameter changes the first-order variance precisely through $I_{\psi\cdot\lambda}$, and Wilks' theorem still gives degrees of freedom equal to the dimension of the restriction on ψ.

Plug-in critical values are valid when the estimated nuisance parameter is consistent and the null approximation is sufficiently uniform. Exact pivots, such as the Student t statistic, are preferable when available. Otherwise one may use a parametric bootstrap, but its validity is itself an asymptotic theorem and can fail in nonregular problems.

0.27 Asymptotic Optimality

An asymptotically valid test is not automatically optimal. In LAN models, optimality is defined by comparison with the Gaussian limit experiment.

0.27.1 Locally Asymptotically Most Powerful Tests

For testing a scalar $H_0 : \theta = \theta_0$ against local upper alternatives, the normalized score test

$$\phi_n = \mathbb{1}\left\{ I(\theta_0)^{-1/2} \Delta_n > z_{1-\alpha} \right\}$$

attains the envelope (0.24.3). No asymptotic level-α test satisfying the regularity conditions can have larger limiting power at the same local direction $h > 0$. Likelihood-ratio and Wald tests have the same first-order local power when their rejection regions are oriented in the same one-sided direction.

With a vector parameter, no test is most powerful in every direction. For a specified direction h, the optimal statistic is proportional to $h^\mathsf{T}\Delta_n$. For an unspecified departure from a smooth null, the likelihood-ratio statistic uses the squared norm of the efficient score in the normal space to the null. This yields rotationally invariant, rather than directionwise, optimality in the limiting Gaussian experiment.

0.27.2 Pitman Efficiency

Suppose two asymptotic level-α tests have limiting local powers of the form

$$1 - \Phi(z_{1-\alpha} - c_j h), \qquad j = 1, 2.$$

Their Pitman asymptotic relative efficiency is

$$\mathrm{ARE}(1,2) = \frac{c_1^2}{c_2^2}.$$

It is interpreted as the limiting ratio of sample sizes required for equal power against nearby alternatives. ARE compares first-order local behavior; it does not determine performance against fixed alternatives or in small samples.

0.27.3 Testing and Confidence Sets

A family of level-α tests can be inverted to form a confidence set:

$$C_n(X) = \{\theta_0 : \text{the test of } H_0 : \theta = \theta_0 \text{ does not reject}\}.$$

If the tests have uniform asymptotic level α, then

$$\liminf_{n\to\infty} \inf_{\theta\in\Theta} P_\theta\{\theta \in C_n(X)\} \geq 1 - \alpha.$$

Pointwise level yields only pointwise coverage. Wald, score, and likelihood-ratio confidence regions arise by inverting their corresponding tests; despite first-order equivalence, their finite-sample shapes can be quite different.

0.28 Nonregular Problems

The standard theory relies on an interior true parameter, differentiability, nonsingular information, identifiable parameters, and a stable support. When these conditions fail, the $n^{-1/2}$ rate, LAN expansion, or chi-squared calibration may be wrong.

0.28.1 Parameters on the Boundary

Suppose a scalar parameter is constrained to $\theta \geq 0$ and the null is $H_0 : \theta = 0$. The null lies on the boundary, so the unrestricted local Gaussian approximation must itself be projected onto a half-line. In a standard one-dimensional case,

$$-2\log\Lambda_n \xrightarrow{d} \tfrac{1}{2}\chi_0^2 + \tfrac{1}{2}\chi_1^2,$$

where χ_0^2 is a point mass at zero. Using an ordinary χ_1^2 critical value is then conservative. More complicated inequality constraints lead to chi-bar-squared mixtures whose weights depend on the local geometry.

The coin model illustrates another boundary failure. At $p_0 = 0$, every observation is zero under the null and the Fisher information formula $1/\{p_0(1-p_0)\}$ is not meaningful. Alternatives on the scale $1/n$, not $1/\sqrt{n}$, can be detectable because

$$P_{p_n}(S=0) = (1-p_n)^n \longrightarrow e^{-c} \quad \text{when } p_n = c/n.$$

The limiting experiment is Poisson rather than Gaussian.

0.28.2 Nonidentification and Singular Information

If a nuisance parameter is present only under the alternative, it cannot be consistently estimated under the null. Mixture-component tests and tests for an unknown change point are common examples. The likelihood-ratio statistic may converge to the supremum of a stochastic process rather than a chi-squared variable.

Similarly, singular Fisher information signals that the quadratic log-likelihood approximation is inadequate. Higher-order terms may determine the rate and limiting distribution. These cases require problem-specific analysis; reporting a standard Wald or Wilks p-value is not justified merely because the sample size is large.

0.28.3 Resampling Calibration

A parametric bootstrap fits the null model, simulates data from the fitted null, and recomputes the statistic to approximate its null distribution. A permutation test instead exploits exchangeability under the null and may be finite-sample exact. The distinction is important:

- permutation validity follows from a group symmetry and can be exact;
- bootstrap validity follows from consistency of an estimated distribution and is generally asymptotic;
- neither method automatically repairs nonidentification or a discontinuous statistic.

The practical asymptotic workflow is therefore: identify the null geometry, derive the statistic's limit under the null and contiguous alternatives, verify whether convergence is uniform over nuisance parameters, and only then choose analytical or resampling critical values. Local power can be compared with the LAN Gaussian envelope when regularity holds; when it does not, the appropriate non-Gaussian limiting experiment determines what optimality can mean.